



MINISTRY OF ENERGY AND MINERAL DEVELOPMENT

LEAST-COST ELECTRICITY GENERATION AND TRANSMISSION PLAN

FY2025/26 – FY2039/40

THEME:

Optimizing Electricity Generation and Transmission
Infrastructure Development for a Sustainable
Electricity Supply Industry

SEPTEMBER 2025





FOREWORD

The Least-Cost Electricity Generation and Transmission Plan (LCGTP) for FY2025/26–FY2039/40 marks a significant milestone in Uganda’s pursuit of a modern, efficient, and sustainable electricity supply industry. As the country advances toward upper middle-income status by 2032, reliable, affordable, and clean energy remains central to stimulating industrial growth, enhancing productivity, and improving the well-being of all Ugandans.

Anchored in Uganda Vision 2040 and the Energy Policy 2023, this Plan provides a clear roadmap for optimizing generation and transmission investments over the next fifteen years. It responds to key sector priorities, including expanding electricity access, strengthening financial and operational sustainability, integrating diverse renewable energy resources, and enhancing security of supply in line with global energy transition trends.

Developed through a consultative, evidence-based process, the LCGTP applies advanced analytical tools such as the Electricity Planning Model (EPM) to evaluate scenarios and identify least-cost, climate-resilient pathways for meeting Uganda’s growing energy demand. It prioritizes strategic interventions that will enhance system adequacy, improve grid reliability, promote regional power trade, and support the decarbonization of the energy mix.

Beyond the technical solutions it proposes, this Plan underscores the importance of coordinated implementation, sustainable financing, and strengthened institutional capacity. It offers Government, sector agencies, and private partners a robust basis for informed decision-making and long-term investment planning.

As the energy landscape evolves, the Ministry remains committed to periodically updating this Plan to incorporate emerging trends, opportunities, and innovations. Working together, we can build a resilient electricity sector that accelerates Uganda’s socio-economic transformation and delivers lasting value for our people.



Hon. Dr. Canon Ruth Nankabirwa Ssentamu
MINISTER OF ENERGY AND MINERAL DEVELOPMENT

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The development of the LCGTP for FY2025/26 – FY2039/40 is a result of extensive collaboration, rigorous modelling, and unwavering dedication from various stakeholders to establish a robust Planning framework for Uganda's electricity generation and transmission infrastructure. We acknowledge the invaluable contributions of individuals and institutions whose efforts made the modelling process possible, leading to the documentation of the results presented in this plan.

I extend my sincere gratitude to the Electricity System Planning and Coordination Committee that provided a platform for spearheading this initiative and providing strategic leadership throughout the development process including technical expertise, guidance, and diligent review of this plan.

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This plan represents a collective vision for Uganda's energy sector. We invite all stakeholders to work together to ensure its successful implementation, ultimately advancing Uganda's aspirations for economic growth, social transformation, and environmental sustainability.



Eng. Irene Bateebe
PERMANENT SECRETARY

UNDERTAKING

The Ministry of Energy and Mineral Development (MEMD), in collaboration with key stakeholders, hereby undertakes to ensure the successful implementation of the Least-Cost Electricity Generation and Transmission Plan (LCGTP) for FY2025/26 – FY2039/40. This plan reflects Uganda's commitment to achieving a reliable, affordable, and sustainable electricity supply to support national development objectives, as outlined in Uganda Vision 2040 and the Energy Policy 2023.

We affirm our dedication to:

1. **Adherence to Policy Objectives:** Aligning all investments and actions with the national goals of ensuring universal electricity access, enhancing grid stability, promoting clean energy, and driving economic transformation.
2. **Collaborative Implementation:** Coordinating with relevant Ministries, Departments, and Agencies (MDAs), private sector stakeholders, and development partners to achieve the objectives outlined in this plan.
3. **Resource Mobilization:** Leveraging both public and private financing to fund the required investments in electricity generation, transmission, and distribution infrastructure.
4. **Periodic Review and Adaptation:** Conducting regular updates and reviews of the LCGTP to accommodate emerging trends, technological advancements, and evolving sectoral demands.
5. **Transparency and Accountability:** Upholding the principles of good governance, ensuring efficient resource utilization, and providing regular progress updates to stakeholders.

We recognize that the successful implementation of this plan is pivotal to Uganda's socio-economic transformation and the attainment of middle-income status. As such, the Ministry commits to mobilizing all necessary resources and efforts to achieve the targets set out in this report.

Through this undertaking, we call upon all stakeholders, both public and private, to play their respective roles in realizing the aspirations of this plan. Together, we can transform Uganda's energy sector into a model of sustainability, reliability, and innovation.

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LIST OF ACRONYMS AND ABBREVIATIONS

BAU	Business as Usual
BESS	Battery Energy Storage System
CAPEX	Capital Expenditure
CBA	Cost-Benefit Analysis
CCGT	Combined Cycle Gas Turbine
CO₂eq	Carbon Dioxide Equivalent
COD	Commercial Operation Date
DRC	Democratic Republic of Congo
DSM	Demand Side Management
EACOP	East African Crude Oil Pipeline
EPM	Electricity Planning Model
ERA	Electricity Regulatory Authority
ESI	Electricity Supply Industry
ESPPC	Electricity System Planning and Coordination Committee
FY	Financial Year
GAMS	General Algebraic Modeling System
GIS	Geographic Information System
GKM	Greater Kampala Metropolitan Area
GoU	Government of Uganda
GWh	Gigawatt hour
HDI	Human Development Index
HFO	Heavy Fuel Oil
IGF	Internally Generated Funds
IPPs	Independent Power Producers
IPTO	Independent Power Transmission Operator
kV	Kilovolt
LCGTP	Least-Cost Generation Transmission Plan
LCOE	Levelized Cost of Electricity
MDA	Ministries, Departments, and Agencies
MEMD	Ministry of Energy and Mineral Development
MoFPED	Ministry of Finance, Planning and Economic Development
MTEF	Medium-Term Expenditure Framework
MW	Megawatt
N-1	Contingency Criterion (loss of a single system element)

NDP III	National Development Plan III
NDP IV	National Development Plan IV
NEMA	National Environmental Management Authority
NPV	Net Present Value
O&M	Operation and Maintenance
OCGT	Open Cycle Gas Turbine
OPEX	Operational Expenditure
PPA	Power Purchase Agreement
PV	Photovoltaic
RAP	Resettlement Action Plan
RES	Renewable Energy Sources
ROR	Run-of-River
SS	Substation
STEI	Science, Technology, Engineering, and Innovation
TCO_{2eq}	Tons of Carbon Dioxide Equivalent
TPP	Thermal Power Plant
UEDCL	Uganda Electricity Distribution Company Limited
UEGCL	Uganda Electricity Generation Company Limited
UETCL	Uganda Electricity Transmission Company Limited
USE	Unserved Energy
VRE	Variable Renewable Energy
WACC	Weighted Average Cost of Capital
YoY	Year on Year

EXECUTIVE SUMMARY

The Least-Cost Electricity Generation and Transmission Plan (LCGTP) for FY2025/26 – FY2039/40 is Uganda’s long-term strategic blueprint to ensure a reliable, affordable, and sustainable electricity supply that supports the country’s industrialization, socio-economic transformation, and transition to clean energy. Anchored in Uganda Vision 2040, the Energy Policy 2023, and aligned with regional and global sustainability goals, the plan adopts a data-driven, multi-scenario approach to optimize generation and transmission investments while meeting rising electricity demand.

Uganda’s electricity demand is expected to increase substantially, driven by rapid urbanization, industrial growth, and a deliberate shift from traditional biomass to modern energy solutions. To address this, the LCGTP models three demand growth scenarios:

Business as Usual (BAU) – 10% annual growth.

Tenfold Growth Strategy – 15.3% annual growth targeting economic transformation.

Energy Policy 2023 Scenario – 18.4% annual growth supporting aggressive decarbonization and industrialization goals by 2040.

Electricity demand is projected to rise from 5,732 GWh in 2023 to between 18,495 GWh and 191,788 GWh by 2040, depending on the scenario, underscoring the need for coordinated investments in generation, transmission, and enabling policies to match Uganda’s long-term development vision.

Using the Electricity Planning Model (EPM), Uganda was divided into seven planning zones i.e. North, West Nile, Central, West, East, Jinja Metro and Greater Kampala Metropolitan Area to evaluate regional demand, generation capacity, transmission bottlenecks, and optimal inter-zonal power transfers. The model integrates renewable energy potential, seasonal load variations, cross-border electricity trade, ensuring the plan reflects both technical and economic realities.

Key Findings

- **Generation Supply strategy:** Hydropower anchors near-term least-cost supply; nuclear becomes a strategic long-term base load in high-growth paths, complemented by accelerated solar, geothermal, storage, and targeted demand-side measures.
- **Transmission Requirements:** Strategic reinforcements are needed, especially the North-GKM and Jinja Metro-East corridors, to meet N-1 and scale inter-zonal transfer capacity.
- **System flexibility:** Hydro and gas provide peaking/balancing; BESS enables higher VRE penetration; spinning reserve requirements rise from 100 MW (to 2030) to 1,400 MW thereafter.
- **Emissions Trajectory & Transition:** Electricity sector emissions range from 0.38 MtCO₂ (BAU) to 14.96 MtCO₂ (Energy Policy). While BAU and Tenfold scenarios remain within the ETP-2030 target, the Energy Policy path requires stronger mitigation through efficiency, cleaner fuels, and carbon management to balance rapid expansion with decarbonization goals.
- **Demand-Side Strategies:** Embedded generation, distributed resources, mini-grids, and load management offer cost-effective ways to reshape the load curve, reduce peak demand, and defer large-scale investments.
- **Nuclear Potential:** Nuclear power provides a strategic pathway to enhance energy security, reduce long-term costs, and diversify the supply mix when paired with careful planning and financing strategies.

- **Regional Trade:** Strengthening participation in the Eastern Africa Power Pool (EAPP) and bilateral interconnections will diversify supply, create export opportunities, and provide a buffer against demand uncertainty.

Total CAPEX Estimates

- BAU Scenario: USD 11.1 billion
- Tenfold Growth Strategy: USD 130.4 billion
- Energy Policy 2023 Scenario: USD 269.2 billion

Funding will be mobilized through a blend of public-private partnerships (PPPs), concessional loans, green bonds, climate finance, and private sector investments

Recommendations

- Prioritize development of projects based on LCGTP insights.
- Comprehensive feasibility studies and full development need to be undertaken for all large power projects.
- Urgent need to operationalize and capitalize the electricity development fund (EDF).
- Fast-track development of additional base load sources.
- Screen large load customers to manage grid and tariff risks.
- Regulate adoption of intermittent generation technologies to safeguard grid stability.
- Develop a national grid flexibility and resilience strategy and plan.

Conclusion

The Least-Cost Electricity Generation and Transmission Plan (LCGTP) is more than a technical blueprint; it is Uganda's integrated energy transition strategy. It ensures that the power sector underpins inclusive economic growth, energy security, regional competitiveness, and environmental sustainability.

The Plan emphasizes not only the deployment of least-cost generation and transmission infrastructure, but also the need for:

- Coordinated and strengthened governance, through operationalization of a central implementation body and enhanced planning coordination via the ESPCC.
- Sustainable financing mechanisms, including through the capitalization and operationalization of the Electricity Development Fund (EDF).
- Accelerated development of diversified base load capacity, including hydro, geothermal, solar, and nuclear projects.
- Smart integration of Variable Renewable Energy (VRE), through a national flexibility and resilience strategy.
- Continuous system planning updates, ensuring the sector remains adaptive to evolving technology, market dynamics, and climate goals.

With robust monitoring, coordinated implementation, sustained financing, and strong stakeholder collaboration, the LCGTP will be pivotal in lighting Uganda's path to middle-income status, universal electricity access, and climate resilience by 2040 and beyond.



1. INTRODUCTION

1.1 BACKGROUND

The Uganda Vision 2040 aims at transforming the Ugandan society from a peasant to a modern and prosperous country by 2040. Achieving this transformational goal will depend on the country's capacity to strengthen the fundamentals including: infrastructure (energy, transport, water, oil and gas, and ICT); Science, Technology, Engineering and Innovation (STEI); land use and management; urbanization; human resources; and peace, security and defence.

Energy, and in particular electricity, is the key driver of socio-economic transformation of a nation and for Uganda to shift from a peasantry to an industrialized and largely urban society, it must be propelled by electricity as a form of modern energy. Affordable, reliable, clean and modern energy services (sustainable energy) facilitate industrialization, value addition, competitiveness, and improved quality of life. Access to affordable, reliable, and clean energy enhances commercial, industrial and agricultural productivity, supports technological advancements, and stimulates economic growth by enabling efficient business operations. It facilitates educational, health and quality water supply services, contributing to human capital development for socioeconomic transformation. Energy infrastructure creates job opportunities and fosters investment thereby reducing poverty and promoting social and gender equity. Sustainable energy enables modern communication services and safety and security, mitigates environmental impacts, and fosters resilience against climate change, which is crucial for long-term socioeconomic stability.

Uganda is one of the fastest growing economies in sub-Saharan Africa, at an average rate of 4.5% annually since 2010 as the country continues its path towards higher economic development and industrialization. Energy is a critical enabler for improving the standards of living of citizens of any country as is evident from the correlation between per capita electricity consumption and the Human Development Index (HDI).

Providing secure, affordable and sustainable energy to all is an important policy priority in Uganda. The Uganda Vision 2040 states: "Ugandans aspire to have access to clean, affordable and reliable energy sources to facilitate industrialization". Uganda Vision 2040 projections indicate that Uganda will graduate into an upper middle-income category by 2032 and become a first-world country in the next fifty years. This commitment was further reaffirmed in the third National Development Plan (NDP III) (2020/2021 to 2024/2025), which prioritized energy as critical for Uganda's aspirations for Vision 2040 and the attainment of upper middle-income status.

Section 17A of the Electricity Act 1999, cap 157, mandates the minister, in consultation with relevant stakeholders, to develop and publish an Energy Policy and Energy Plan to give direction and form a basis for investment in the power sector aimed at a sustainable demand/supply balance.

The Government of Uganda has, over the years, developed plans to set the direction for the Electricity Supply Industry (ESI) in Uganda. Previous energy planning in Uganda has emphasized supply-side issues such as increasing generation and attracting more private investors, especially for commercial energy sources. Demand-side issues,

such as subsidies to counteract high end-user power tariffs and meeting energy demand by increasing area coverage, have received less attention. This approach has tended to favour the urban population, which is the major user of commercial fuels, while marginalizing the energy needs of the majority of the population in rural areas, which depends mainly on biomass.

A sustainable energy sector plan integrates economic, social and environmental objectives to improve the well-being of the current generation whilst safeguarding the welfare of future generations.

To address the aspirations of the Energy Transition Plan (ETP) 2023 and Energy Policy 2023, the Government of Uganda has been supported by the World Bank to develop a Least Cost Generation Transmission Plan (LCGTP) for Uganda.

Currently, access to electricity, as measured by the proportion of households connected to either grid or off-grid technology has increased to 60% in FY2023/24 from 24% in FY2017/18. This is mainly driven by the increased uptake of off-grid technologies (38%) such as mini-grids, stand-alone solar systems, and solar home kits. As such, electricity consumption per capita has increased to 218 kWh from 100 kWh in FY2017/18. However, this is below the National electrification and decarbonization aspirations as set out in the ETP.

Furthermore, Uganda still has high proportions of suppressed demand estimated at 2,420 MW in 2022 largely due to limited inter and intra-sectoral investment planning and inadequacies in the electricity distribution and transmission network, which results in the reliability and quality of power supplied being poor. The inadequacies include ageing and dilapidated electricity distribution, inadequate transmission infrastructure, vandalism, inadequate technical network monitoring, and limited reinvestment in operations and maintenance, among other operational challenges. The above issues have constrained the evacuation of power to key load centers such as cities, urban centers and some industrial parks, thereby limiting the exploitation of potential markets and financial viability of the electricity supply investments.

The installed electricity generation capacity has increased over the years from 984 MW in FY2017/18 to 2005 MW in FY2023/24 following completion of the flagship 600 MW Karuma hydro power plant (HPP), the 183 MW Isimba HPP, and several privately owned power plants contributing over 200 MW. However, this generation capacity is inadequate to address the existing suppressed demand. The electricity demand is exacerbated by the Country's 10-fold growth aspirations, including industrialization, value-addition, e-mobility, STI and transportation.

Additionally, while the electricity generation mix has been diversified to include other technologies such as grid-connected solar PV and cogeneration, a large proportion is contributed from hydro, which is collocated along the Nile cascade.

The country's electricity system has limited deployment of digital and novel innovations such as modern energy storage systems to optimize expansion, operation and maintenance. Some hydro generating stations are aged and as such require detailed condition based technical assessments and rehabilitations, while other power plants shall necessitate hybridization with solar PV and gas to meet the greening aspirations of the Country. The transmission and distribution networks have insufficient smart grid technologies, automatic generation control, live line maintenance, remote transformer monitoring, data concentrator units, net metering, and grid forming technologies among others. These technologies are not only critical

in improving the quality of power supply but also reduction of losses.

The use of traditional biomass for cooking has reduced, however, it remains the major source of cooking energy. There has been a reduction in the use of traditional biomass for cooking to 75% in FY2022/23 from 85% in FY2017/18. However, the technologies used are rudimentary and highly inefficient, mostly characterized by open fires and basic stoves. This has led to dire consequences for health, gender inequality, climate, and the environment. Unsustainable use of biomass resources continues to aggravate energy poverty. Conversion of this demand to cleaner and modern forms is expected to contribute to at least 25,217MW of new electricity demand in the Energy policy scenario.

There have been efforts to strengthen the policy, legal, institutional, and coordination framework, however, gaps exist which hinder effective oversight, coordination and operation of the electricity supply industry. In addition to the infrastructure deficiencies, the energy industry's institutional framework is characterized by institutional gaps and capacity deficiencies within key agencies, leading to slow implementation, investment, research, and innovation. Furthermore, weak coordination among the key sector players impedes effective investment in the industry.

The development of the Least Cost Generation and Transmission Plan (LCGTP) is a step towards realizing the envisaged harmonization of the Electricity Supply Industry planning function. Prior to the Amendment of the Electricity Act 1999, planning in the sector has entirely been adhoc with a segmented approach to planning. The Electricity Regulatory Authority created the System Planning and Coordination Committee (SPCC) that brought together generation, transmission and distribution entities. However, the SPCC largely relied on the segment plan which necessitated harmonization as envisioned in the Amended Act. The ERA has, over the years, developed and updated the Least Cost Electricity Expansion Plan (LCEEP) on an annual basis while UETCL developed the Grid Development Plan (GDP). To align and coordinate planning, the mandate has been given to the Minister of Energy and Mineral Development, under which the LCGTP has been developed with key stakeholder involvement.

Currently, under the MEMD as enshrined in the Electricity Act as amended in 2022, a sector committee, the Electricity System Planning and Coordination Committee (ESPCC) has been instituted to bring together all sector players across the electricity value chain to plan for generation and transmission expansion, thus the Least Cost Generation and Transmission Plan (LCGTP). This is aligned with the Energy Transition Plan (ETP) that stipulates the roadmap for modernizing and developing Uganda's energy sector, the Grid Development Plan focusing on expanding and strengthening the electricity grid to meet the growing demand for electricity and ensure reliable supply, and the Integrated Energy Resource Master Plan. The LCGT plan has been aligned to the key policy documents with a focus on the NDP IV, the Energy Policy 2023, the existing Grid Development Plan and the East African Community Vision 2050.

This Least-Cost Electricity Generation and Transmission Plan defines the optimal electricity generation and transmission system expansion requirements over the medium-term (FY2025/26 – FY2039/40). This will be reviewed and updated regularly (annually or every 2 years) and feeds into the planned Integrated Energy Resource Master Plan for Uganda's Energy Sector.

The Integrated Energy Resource Master Plan identifies an optimal mix of resources,

transmission, and distribution that meet the long-term (FY2025/26 – FY2049/50) consumer energy needs in Uganda efficiently and reliably at the lowest reasonable cost while transitioning to a cleaner and greener energy source. The Integrated Energy Resource Planning integrates resource options for the whole energy sector (both supply and demand sides) to develop a long-term strategy for meeting the sector needs over the planning horizon.

1.2 Governance and Institutional Arrangements for the Electricity Supply Industry (ESI)

The Energy Policy for Uganda 2023 aims to ensure a sustainable, adequate, affordable, competitive, secure, and reliable supply of energy at the least cost, geared to meet national and county needs while protecting and conserving the environment.

The Electricity (Amendment) Act 2022 mandates the Minister to develop in a consultative manner, an energy policy and attendant generation, transmission and distribution plans. The energy plans developed by the Minister should consider the national energy policy, serve as a guide for energy infrastructure investments, consider all viable energy supply options, and guide the selection of appropriate technology to meet energy demand.

The Ministry, through the multi-sectoral Electricity System Planning and Coordination Committee (ESPCC), facilitates the Minister in the execution of the planning mandate as outlined in the Electricity (Amendment) Act 2022 and in strategically overseeing the development, harmonizing, coordination and implementation of the Electricity Supply Industry (ESI) Strategies and Investment plans. Specifically, the Committee undertakes the following:

- i. Operationalize and coordinate comprehensive ESI planning encompassing generation, transmission, distribution, energy access initiatives and energy efficiency as well as promoting penetration of renewables.
- ii. Oversee the operational efficiency of the grid, identify constraints, and propose effective interventions to mitigate and address these limitations.
- iii. Assess the capacity and reliability of the existing ESI infrastructure and ensure that the capacity and reliability of the grid system are not compromised.
- iv. Coordinate policy and strategy formulation in the ESI
- v. Undertake risk analysis for the investment and operations & maintenance plans.

The Committee is constituted of the following permanent member institutions:

- i. The Ministry of Energy and Mineral Development is mandated to Establish, Promote the Development, Strategically Manage and Safeguard the Rational and Sustainable Exploitation and Utilization of Energy and Mineral Resources for Social and Economic Development.
- ii. The Electricity Regulatory Authority, which is a Statutory Body that regulates the Generation, Transmission, Distribution, Sale, Export, and Import of Electrical Energy in Uganda.
- iii. The Uganda Electricity Generation Company Ltd (UEGCL) that is mandated to establish, acquire, maintain and operate electricity generation facilities and to promote Research and Development in the Electricity generation sector while running on sound business principles.

- iv. The Uganda Electricity Transmission Company Ltd (UETCL) that manages the high voltage electricity transmission infrastructure in the country.
- v. Uganda Electricity Distribution Company Ltd (UEDCL) that manages the Electricity Distribution Network in the country.
- vi. The National Planning Authority.
- vii. Uganda Investment Authority.

1.2.1 Least Cost Generation and Transmission Plan

The Least Cost Generation and Transmission plan (LCGTP) is intended to provide policy makers and other stakeholders in the electricity supply industry with information about the short, medium and long-term demand outlook in Uganda, and the required investments in the transmission network and generation capacity to meet the projected demand.

Specifically, the LCGTP undertakes to achieve the following:

- b. To determine the optimal least-cost generation and transmission mix required to meet future electricity demand while ensuring security of supply and achieving national access targets, industrial growth, and economic development.
- c. To determine the investment requirements that prioritizes least-cost and high-impact projects for generation and transmission infrastructure development, ensuring alignment with national energy policies and long-term targets.
- d. To provide data-driven insights and recommendations that inform policy decisions on electricity generation, transmission, and overall sector development, ensuring alignment with national goals, regulatory frameworks, and international best practices.

1.2.2 The Plan Formulation Approach

Formulation of the 1st LCGTP adopted an all-inclusive stakeholder consultative approach. The process was guided by the World Bank's Electricity Planning Model (EPM), a robust tool powered by the General Algebraic Modeling System (GAMS) [1]. The Model facilitated scenario optimization, integration of ancillary services and environmental factors into the planning process. The use of these advanced tools ensured a data-driven and technically sound approach to achieving least-cost solutions.

To ensure that the Plan is aligned with existing strategies and policies, the process involved a comprehensive review of key industry documents such as the Energy Policy for Uganda, 2023; the Energy Transition Plan for Uganda; the UETCL Grid Development Plan and the National Electricity Generation Strategy, among other technical reports and policy documents.

1.3 Scenarios

The Plan is based on three scenarios:

- i. Business As Usual+: This scenario assesses Uganda's electricity supply industry development trajectory at a demand growth rate of 10%. This is based on

addressing electricity transmission constraints to key load centers such as West Nile and optimization of the already deployed infrastructure in key industrial parks and implementation of the electricity connections policy.

- ii. Tenfold growth strategy scenario: This scenario evaluates the ESI development trajectory with a 15% electricity demand growth rate following the Government's Strategy to grow the economy tenfold in the next 15 years, starting with double-digit in the NDPIV implementation period. The 10-fold growth strategy is anchored by the country's desire and ambition for socio-economic transformation in the current trend of ATMs, science, technology & innovations, and globalization.
- iii. Energy policy 2023 growth scenario. This scenario assesses a development trajectory with an electricity demand of 18.4%. The demand for electricity is largely driven by the need to transition from biomass that currently accounts for 87.8% of the energy mix by 50%; Uganda's Industrialization Agenda; and the overall transition to clean energy for several services.

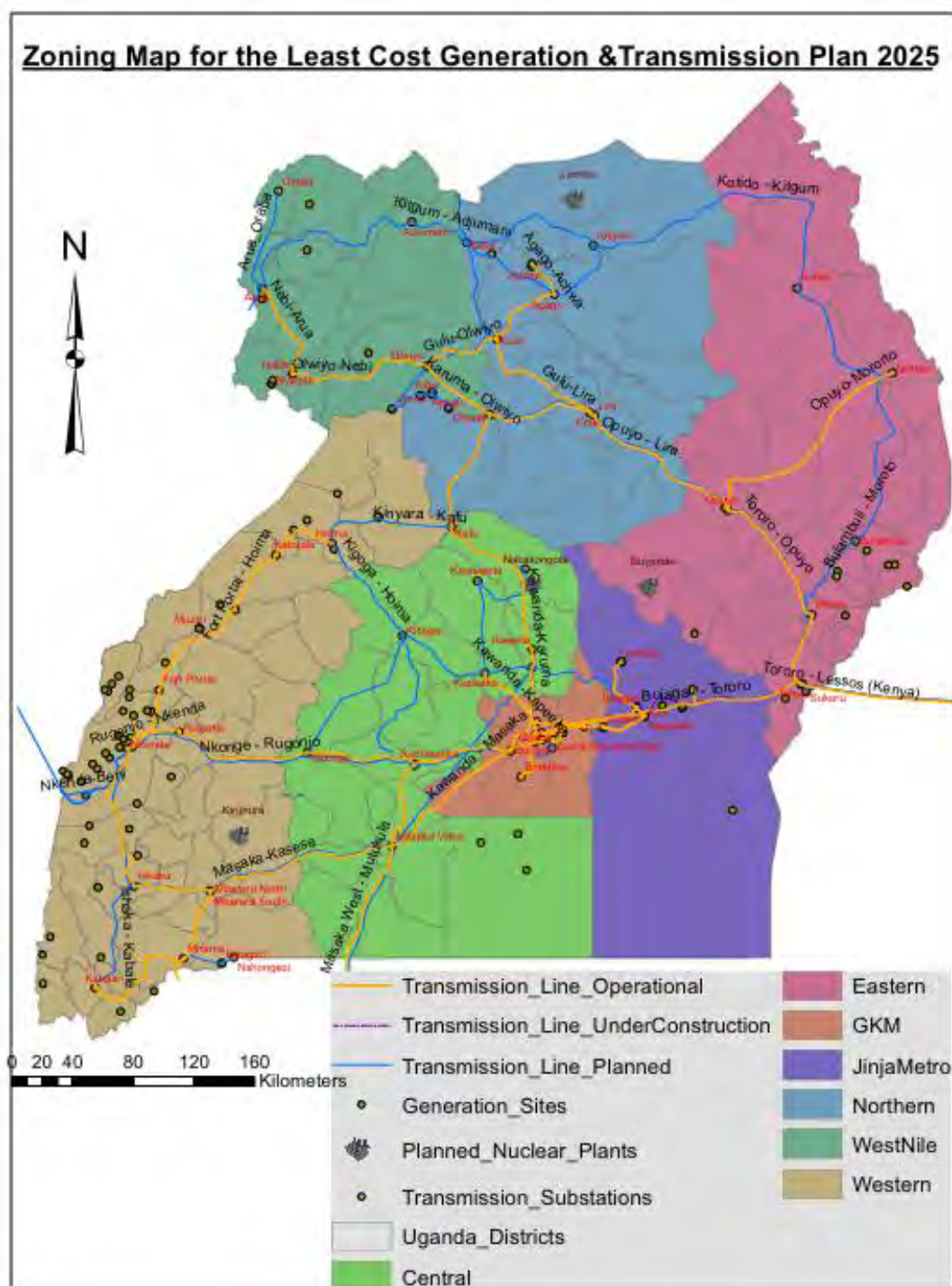


1.4 Transmission zoning

1.4.1 Zoning map

The map in Figure 1-1 below illustrates the division of Uganda's power system into seven strategic zones to guide transmission expansion investment decisions. These zones include the Greater Kampala Metropolitan Area (GKM), Jinja Metro, Central, Eastern, Western, Northern, and West Nile. This zonal approach enables the optimization of electricity flows between regions by considering transmission capacities, regional demand, and available generation resources.

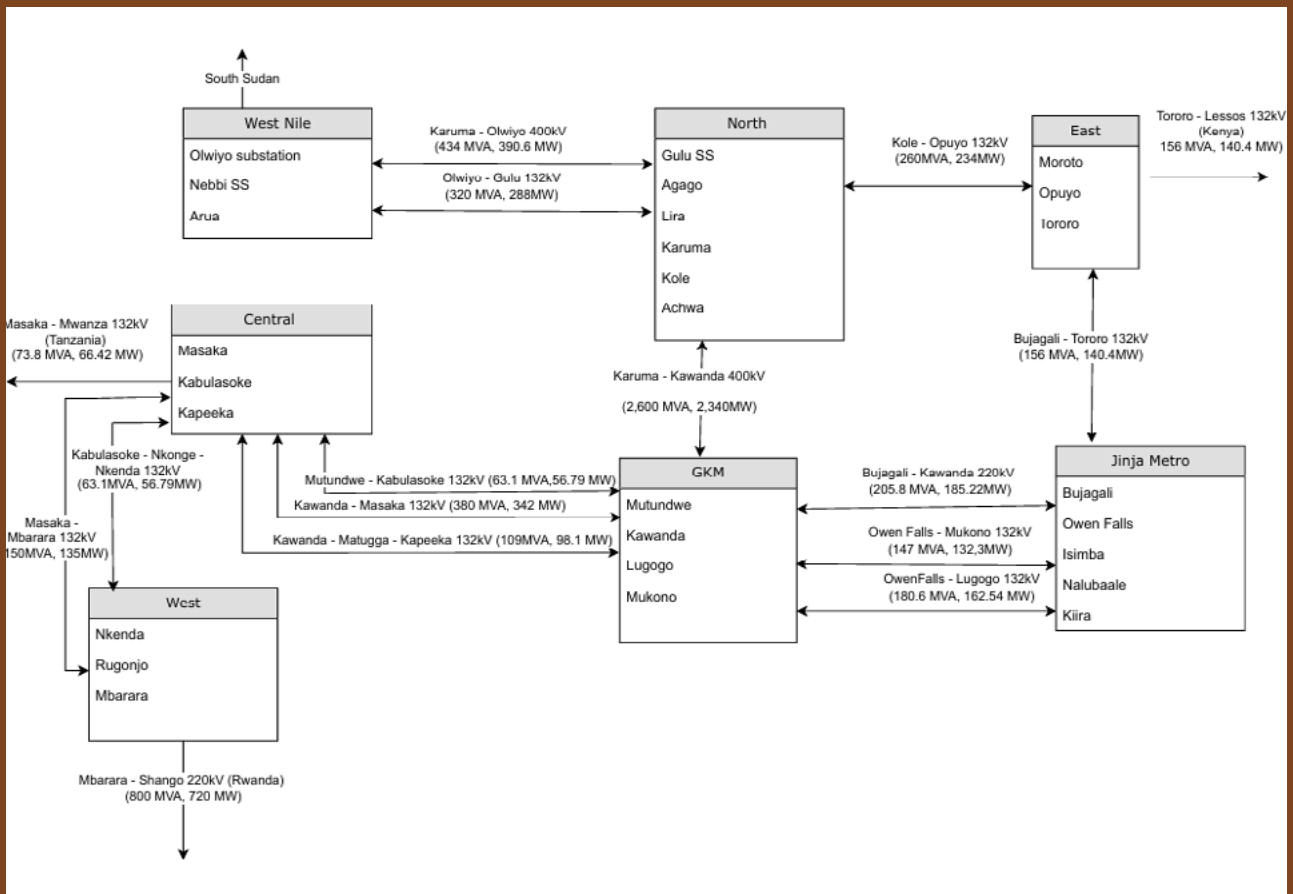
Figure 11: Zoning Map for the Least Cost Generation Transmission Plan



1.4.2 Interzonal Transfer Capacities

The diagram in Figure 1-2 below shows the transmission interconnections among Uganda's seven strategic power zones under the Least Cost Generation and Transmission Plan. It outlines key 132 kV, 220 kV, and 400 kV lines, along with substations, power transfer capacities, and inter-regional energy flows. The current network enables the evacuation of power from generation hubs like Karuma and Owen Falls to high-demand zones such as GKM, while also linking Uganda's grid to neighboring countries including Kenya, Tanzania, Rwanda, and South Sudan.

Figure 12: Schematic diagram showing the net transfer capacity across zones



While Uganda's current inter-zonal transmission network enables power transfers across the seven planning zones, the system remains vulnerable and lacks redundancy in several critical corridors. Notably, the North-GKM corridor, which depends solely on the Karuma-Kawanda 400 kV line, is a major point of concern. As the primary evacuation route for major hydropower power plants (e.g. Karuma, Bujagali), any failure along this path significantly constrains power flow into the central and east load centers. This violates the N-1 contingency criterion, which requires the system to withstand the loss of any single element without causing widespread service interruption.

In addition to the North-GKM corridor, the North-East and East-Jinja Metro corridors also lack adequate parallel transfer paths and exhibit low system redundancy, especially given the projected increase in regional demand and power flow. These bottlenecks expose the system to congestion, curtailments, or isolation of entire zones during outages or maintenance periods. Similarly, weaknesses in the West Nile-North and Central-West corridors further illustrate the need for strategic reinforcement.

1.5 Past Electricity Demand Growth

The Table 1-1 below shows the historical electricity energy and peak demand recorded of the Ugandan power system for the past five years. It's observed that the grid connected peak demand of the Uganda power system has increased at an average of 50MW annually over the period 2018 to 2022 as portrayed in the table below.

Table 11: Historical Trend of Energy sales and Power peak demand

Year	Domestic Peak Demand (MW)	Peak System demand (MW)	Domestic Energy Sales to Distribution (GWh)	Domestic Energy Sales + Export
2018	600	651	4,102	4,371
2019	641	729	4,171	4,449
2020	689	705	4,536	4,706
2021	728	804	5,055	5,508
2022	796	825	5,410	5,823

Historically, the electricity system energy sales have grown at an average of 7.6% Year on Year (YoY) for the last five years.

1.6 Electricity access

Electricity access currently stands at 60% (22% - grid and 38% - off grid). This is mainly driven by increased uptake of off-grid technologies (38%) such as mini-grids, stand-alone solar systems and solar home kits. As such, electricity consumption per capita has increased to 218kWh from 100kWh in FY2017/18. However, this is below what is required to significantly reduce poverty and enhance productive use. Of the 2,430,489 electricity customers, 0.20% are industrial, 4.01% commercial and 95.78% domestic.

1.7 Current electricity generation capacity

1.7.1 Generation Capacity

Uganda has, over the last 2 decades, developed over 1,600MW which is mainly dominated by run of river.

The Table 1-2 below shows the installed generation capacity over the past years.
Table 1-2: Installed generation capacity over the past years

Years	2024	2020	2015	2010	2005	2000
Capacity (MW)	2,051.4	1,268	895.5	690.4	444.9	385.0

The graph in Figure 1.3 illustrates the growth of Uganda's installed electricity generation capacity from 1950 to 2024. The data shows a gradual increase in capacity from just 5 MW in 1950 to 2,051.4 MW by the end of 2024.

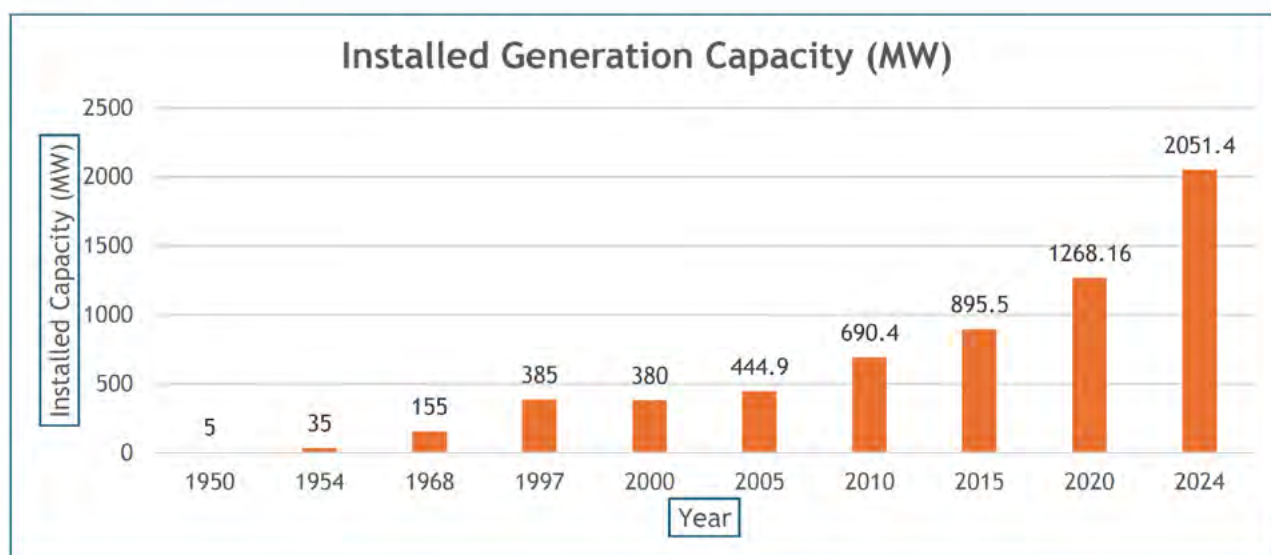


Figure 1-3: Uganda's installed generation capacity from 1950 to date.

The installed capacity stands at 2,051.4 MW as at the end of the year 2024, of which 11.6 MW relates to Off-Grid systems and 2,039.8 MW is on the National Grid.

1.7.2 Installed Electricity Generation Capacity Mix

Uganda's electricity generation portfolio remains dominated by hydropower, reflecting the country's historical reliance on the Nile basin for large-scale generation. As of 2024, the total installed generation capacity stands at 2,051.4 MW, comprising both grid-connected and off-grid systems.

Table 1-3 below provides a breakdown of this capacity by technology, highlighting the relative contributions of large hydros, mini hydros, solar PV, thermal, cogeneration, and biomass.

Table 1-3: Installed generation capacity by Technology, 2024.

Technol-ogy	Large Hydros	Mini Hydros	Solar PV	Thermal	Cogenera-tion	Biomass	Total
Capacity (MW)	1500.2	215.1	86.7	101.1	147.9	0.4	2051.4

This generation mix dominated by hydropower (over 83%) raises important policy considerations regarding system vulnerability to hydrological variability and climate shocks. New technologies such as nuclear power are under consideration for mid-to-long-term deployment to provide firm base load capacity and enhance energy security. At the same time, variable renewable energy (VRE) sources notably solar PV and wind are gaining traction, supported by falling technology costs, global climate commitments, and growing private sector interest.

1.8 TRANSMISSION NETWORK

1.8.1 Status of the transmission network

Uganda's high-voltage transmission network has undergone considerable expansion, enhancing its ability to transmit bulk power from generation sources to distribution networks and regional interconnection points. As of December 2024, the country's total transmission circuit length stands at approximately 5,139.7 kilometers, comprising of 132 kV: 3,496.5 km; 220 kV: 1,002 km; 400 kV: 606 km and 66 kV: 35.2

km.

In tandem, the national grid includes 43 substations with a combined transformation capacity of 6,985.5 MVA, distributed across multiple voltage levels. The system integrates transformation points at 400/220 kV, 400/132 kV, 220/132 kV, 132/33 kV, and 132/11 kV, supported by mobile substations for temporary or emergency deployment.

1.8.2 Current interzonal transmission capacity

Uganda's high-voltage transmission backbone is supported by a network of key interzonal and cross-border transmission corridors. These lines are critical for facilitating regional power flow, ensuring domestic load balancing, and enabling power exports and imports with neighboring countries.

Table 1-4 Existing transmission lines between the zones.

Zone	Transmission Line	MVA	Max. MW	km
West Nile to North	400kV Karuma - Olwiyo	434	391	55
	132kV Olwiyo - Gulu	320	288	57
Central to GKM	132kV Mutundwe - Kabulasoke	63	57	18
	Kawanda - Masaka 220kV	380	342	137
	Kawanda - Matugga - Kapeeka 132kV	109	98	52
Central - West	Kabulasoke - Nkongwe - Nkenda 132kV	63	57	279
	Masaka - Mbarara 132kV	150	135	131
Jinja Metro - East	Bujagali - Tororo 132kV	156	140	122
Jinja Metro - GKM	Bujagali - Kawanda 220kV	206	185	75
	Owen Falls - Mukono - Namanve 132kV	147	132	5
	Owen Falls - Lugogo 132kV	181	163	70
North - GKM	Karuma - Kawanda 400kV	2,600	2,340	248
North - East	Lira - Opuyo-Tororo 132kV	260	234	263
East - Kenya	Tororo - Lessos 132kV	156	140	27
West - Rwanda	Mbarara - Mirama - Shango 220kV	800	720	66
Central - Tanzania	Masaka - Mwanza 132kV	74	66	85
West Nile - South Sudan	Olwiyo - Juba 400kV	2600	N/A	150

The Table 1-4 above outlines the major corridors, including line voltages, transformation capacities (MVA), maximum transfer limits (MW), and line lengths (km). Strategic interconnectors like Karuma-Kawanda 400 kV, Karuma-Olwiyo 400 kV, and Bujagali-Kawanda 220 kV serve as the primary channels for bulk power movement from generation-rich regions to major consumption centers.

2 PLANNING METHODOLOGY

2.1 GUIDING PRINCIPLES OF LEAST-COST PLANNING METHODOLOGY

The Least-Cost Planning Methodology is designed to ensure that electricity generation and transmission investments are optimized to deliver reliable, affordable, and sustainable power. This approach considers economic, technical, and policy factors to achieve the most efficient expansion of the power sector. The key guiding principles include:

- Explicitly defining the total system costs, with a clear breakdown of the components being minimized, including both CAPEX and OPEX.
- Identifying and adhering to key constraints, including technical limitations and policy requirements.
- Implementing a multi-year intertemporal optimization approach to ensure long-term efficiency and sustainability.

2.2 OVERVIEW OF THE ELECTRICITY PLANNING MODEL

The World Bank's Electricity Planning Model (EPM) is a long-term, multi-year, multi-zone capacity expansion and dispatch model. The objective of the model is to minimize the sum of fixed (including annualized capital costs) and variable generation costs (discounted for time) for all zones and all years considered, subject to:

- Demand equals the sum of generation and non-served energy,
- Available capacity is existing capacity plus new capacity minus retired capacity,
- Generation does not exceed the max and min output limits of the units,
- Generation is constrained by ramping limits,
- Spinning reserves are committed every hour to compensate for contingencies and VRE forecasting errors,
- renewable generation constrained by wind and solar hourly availability,
- Excess energy can be stored in storage units to be released later or traded between the other zones,
- Transmission network topology and transmission line thermal limits.
- Capacity reserve margin of 10%, and
- Availability of the existing power plants is constrained based on historical generation.

EPM is used mostly to perform least cost expansion plans as well as dispatch analyses. Developed by ESMAP since 2015, EPM has been used to study dozens of countries. EPM uses a core mixed integer programming multi-zone model, implemented in the General Algebraic Modelling System (GAMS), to minimize the total discounted costs of Capex and Opex.¹ The model optimizes the generation (and transmission if included) expansion over future years of the modelled time period, as well as the underlying dispatch of generation and power flows on transmission lines for existing and new transmission assets. In addition to the core decision variables on generation and transmission capacity addition, dispatch, and flow, the model also co-optimises spinning reserve provision from the generators. EPM has been implemented for 100 developing countries over the last 10 years and is being used for many types of analyses.

The model is derived based on the following assumptions:

1. Market participants are not strategic and they behave in a perfectly competitive manner, i.e., the power plant owners submit their true costs as bids.
 2. The forecasted demand is considered perfectly inelastic, which implies that the maximization of the social welfare can be replaced by minimization of the system cost.
 3. The trade among regions/countries is economically efficient (optimal), which translates to a single objective of minimization of cost for the whole region/country.
 4. The pricing is efficient and does not provide incentives to market participants to deviate from the optimal behavior.
- Identify investment needs (generation mostly) to meet the demand under various policy scenarios
 - Determine where and by how much renewable resources should be deployed to maximize their value to the system.
 - Determine the optimal capacity additions over time to complement renewable generation accounting for existing generating units, energy storage, demand-side response and/or a carbon constraint.
 - Determine the optimal retirement schedule of the existing units over time.
 - Assess the utilization of the transmission lines (important to design trade contracts).
 - Determine the optimal transmission capacity additions.
 - Determine hourly electricity prices with trade for different countries and zones, which is essential to value the energy traded.
 - Determine the operations and expansion cost to the consumer.
 - Determine the impact of different market conditions (e.g., fuel prices, fuel subsidies, carbon limits, etc.) and technology cost assumptions on the optimal capacity expansion plan and the optimal energy mix.
 - Determine the cost of implementing specific environmental policies: renewable portfolio standard, cap on carbon emissions, tax on carbon emissions, and carbon emissions rate.

2.2.1 Note on Regional Trade Modelling:

While the Electricity Planning Model (EPM) is capable of modelling cross-border power trade, regional electricity trade was not explicitly modeled in this version of the Plan. This decision was based on the limited availability of reliable long-term data on neighboring countries' generation expansion plans, demand trajectories, and firm trade agreements. However, Uganda's participation in the East African Power Pool (EAPP) and ongoing cross-border infrastructure projects have been considered qualitatively in scenario design and transmission planning. Future iterations of the Plan will aim to incorporate dynamic regional trade modelling as more coordinated regional data becomes available.

2.3 IMPLEMENTATION OF THE ELECTRICITY PLANNING MODEL

2.3.1 Modelling Framework and Planning Approach

The modelling approach employed in this Plan is grounded in the Electricity Planning Model (EPM), a linear optimization framework developed by the World Bank and implemented in the General Algebraic Modelling System (GAMS). The model determines the least-cost mix of generation and transmission investments required to meet projected electricity demand over the planning horizon, while satisfying a comprehensive set of technical, economic, and policy constraints.

The EPM operates on hourly dispatch resolution within representative time slices, capturing daily and seasonal variations in load, renewable energy profiles, and hydro inflows. It ensures that all energy and capacity requirements are met while respecting system adequacy, spinning reserves, operational flexibility, and emissions limits. National policy objectives such as renewable energy targets and technology-specific mandates are directly incorporated as binding constraints within the optimization.

Crucially, the model integrates both generation and transmission planning, optimizing power flows across zones based on existing and candidate interconnection capacities. It identifies network bottlenecks and endogenously determines when and where to invest in grid reinforcements or expansions to minimize overall system cost. Reliability is further reinforced through complementary contingency (N-1) analysis using PSSE software detailed in Grid Development Plan, ensuring that the system can withstand single-point failures without major service interruptions.

This integrated, spatially-aware framework enables consistent and transparent planning across the electricity value chain. It supports cost-effective infrastructure deployment, promotes grid reliability, and enhances regional energy trade, positioning the system to accommodate future challenges such as high VRE integration, climate variability, and evolving demand patterns.

2.3.2 Multi-zonal Model with Transmission Decisions across Zones Being Optimized

To support spatially coherent system planning, the country was divided into seven strategic zones, namely Greater Kampala Metropolitan Area (GKM), Central, Jinja Metro, Eastern, Northern, Western, and West Nile, based on existing transmission infrastructure, demand clusters, and generation sources. This zonal breakdown enables the model to capture the regional diversity in demand growth, resource availability, and grid constraints, while optimizing generation siting and power transfers across the network.

2.3.3 Resolution

The model operates on an hourly time resolution, but to reduce computational complexity while preserving key system dynamics, each month is represented by a set of two representative days. These are carefully selected to capture the distinct characteristics of electricity demand and generation variability within that month. Specifically, the model includes:

- An average day, reflecting the typical daily load profile for the majority of the month, and
- A peak day, representing the day that contains the month's highest demand hour.

This representative-day approach enables the model to effectively simulate seasonal variations in electricity demand, renewable energy generation profiles (e.g., hydropower and solar PV), and fuel availability, without the computational burden of simulating every individual day.

By incorporating this methodology, the model balances computational efficiency with modelling accuracy, allowing for the clear identification of monthly trends that influence generation dispatch, intraday and seasonal variability of solar, wind and hydro resources, transmission utilization, system costs, and fuel supply risks. This ensures a robust and practical basis for long-term electricity planning.

2.3.4 Ancillary Services Co-Optimized

In addition to optimizing energy production, the model simultaneously co-optimizes the provision of ancillary services, with a particular focus on spinning reserves. This ensures that generation resources not only meet energy demand but also contribute to maintaining system reliability and operational security.

The model allocates available generation capacity between energy supply and ancillary services in a cost-effective and technically feasible manner. By embedding ancillary service requirements within the least-cost dispatch framework, the model accurately reflects the trade-offs between reliability and economic efficiency and identifies the most suitable technologies for providing reserves.

In terms of spinning reserve requirements, the model considers two common planning criteria:

- The capacity of the single largest generating unit on the system,
- Reserve set at 15% of variable renewable generation to account for forecasting errors and output variability as per the EPM methodology.

For this planning horizon:

- Between 2023 and 2030, the spinning reserve requirement was set at 100 MW, corresponding to the capacity of the largest turbine unit at Karuma HPP.
- From 2031 onwards, as larger baseload units (e.g., nuclear or other firm capacity) are expected to come online, the reserve requirement was adjusted to 1,000 MW, in line with the anticipated largest generator unit size.

This approach ensures that the system maintains adequate reserve margins to respond to sudden generation outages or demand surges, thereby enhancing system resilience and frequency stability throughout the planning period.

Table 2-1 Spinning Reserve allocation

Year	Maximum demand (MW)	Largest available unit	Spinning Reserve Units (MW)	Percentage of installed capacity for reserve provision	Power Plants to provide the spinning reserve
2023 - 2029	1,064	100	100	50%	Karuma
				32%	Eskom
2030-2040	22,000 - 52,000	1,400	1,400	70%	Bess
				40%	Central ror com, east ror com, west ror com, east_ror_c and west_ror
				25%	Central diesel
				20%	Tilenga gas tpp, east_gt, west_gt
				10%	Kingfisher gas tpp, east_ccgt, west_ccgt
				5%	Scoul bagassebio, central_unergy_bio, central_bio, central_nakasongola_npp, East_buyende_npp, kamuli_sugar_bagasse_bio, pramukh__bagasse_bio, north_pece_bio, north_ror - 40, north_lamor_npp, kabale_peat_bio, kyoga_peat_bio, west_biomass, west_kiruhura_npp.

The Table 2-1 above outlines the spinning reserve requirements and the corresponding power plants designated to provide this reserve in Uganda's electricity grid.

The reserve provision strategy progressively broadens over time, shifting from reliance on a few large hydro units to a diversified reserve portfolio. Post-2030, the model distributes spinning reserve obligations across multiple technologies and regions, enhancing system resilience, flexibility, and redundancy in line with evolving generation dynamics.

The source GAMS code and documentation including the mathematical formulation is available online: <https://github.com/worldbank/EPM>

2.3.5 Carbon Quantity Modeled

To support Uganda's climate commitments under the Energy Transition Plan (ETP) and broader sustainable development goals, the model incorporates a carbon quantification mechanism that reflects the cost of greenhouse gas (GHG) emissions associated with generation. This enables reporting of the carbon footprint of different generation pathways and assessment of trade-offs between cost, emissions, and long-term energy security.

The Table 2-2 below outlines the CO₂ emissions factors used for key fossil fuel types in power generation. These values, expressed in tons per million British thermal units (ton/MMBTU), are applied within the model to estimate the scale of emissions from each technology. Fuels with higher emission intensities, such as diesel and HFO, therefore register higher reported emissions in the results, but these do not directly affect the relative performance or optimization of generation options in the model.

Table 2-2: Emissions Factor (ton/MMBTU)

Fuel Type	CO ₂ Emissions Factor (ton/MMBTU)
CCGT	0.05306
OCGT	0.05306
Diesel	0.07325
HFO	0.07510







3 INPUT DATA AND ASSUMPTIONS

3.1 SYSTEM LEVEL ASSUMPTIONS

Table 3-1 below presents the key financial and technical parameters that form the foundation of energy project evaluation within this Plan, particularly for electricity generation and transmission investments. These assumptions are integral to the EPM's optimization process, influencing both investment decisions and scenario outcomes. They ensure consistency in assessing system costs, reliability, and alignment with long-term planning and policy objectives.

Table 3-1: Key system level GAMs model assumption

Parameter	Value
Weighted Average Cost of Capital (WACC), %	11%
Discount rate, %	11%
Cost of unserved energy per MWh, \$	2,500
Cost of planning reserve shortfall per MW, \$	100,000
Spinning Reserve Voll per MWh, \$	50
Cost of curtailment per MWh, \$	0.5
Cost of Climate backstop techno per ton CO ₂ , \$	300
Capacity reserve margin	10%
Contribution to the system requirement for firm capacity - PV	3%
Contribution to the system requirement for firm capacity - Wind	25%

Source: Ministry of Energy and Mineral Development

3.2 ELECTRICITY DEMAND FORECASTING

3.2.1 Demand Forecasting Methodology

3.2.1.1 General approach

The demand forecast applied in this Plan adopts a bottom-up approach, consistent with national integrated resource planning practice. Electricity demand was disaggregated by customer category (Domestic, Commercial, Medium Industrial, Large Industrial, Street Lighting and additional categories like step loads), and

projections were developed based on historical consumption trends, growth drivers, and sector-specific assumptions.

Two key techniques were used:

- Sector-based forecasting, which uses historical sales and economic indicators (e.g., GDP, urbanization, electrification targets) to project future energy consumption by category.
- Load shape modeling, where typical hourly load profiles were applied to each customer category using representative average and peak days per month, capturing both daily and seasonal variations.

The methodology also accounts for future electrification efforts, and new load additions such as oil and gas infrastructure, electric mobility, and electric cooking. Adjustments were made for biomass substitution and distributed energy uptake, in line with national energy access and transition objectives.

This disaggregated, time-resolved forecast provides the hourly demand input for the Energy Planning Model (EPM), enabling cost-reflective optimization of generation and transmission investments over the 2024-2040 horizon.

3.2.1.2 Drivers of the projected demand

The demand forecast is underpinned by several key drivers that shape both short- and long-term electricity consumption trends across Uganda:

- i. **Demographics:** Population growth, rural-to-urban migration, and rising household electrification directly influence domestic consumption. Urbanization, in particular, drives increased demand for commercial services, public infrastructure, and municipal loads such as street lighting.
- ii. **Economic Growth (GDP):** Projected GDP growth plays a central role in determining electricity consumption across households and productive sectors. Growth in income levels and industrial activity translates to higher demand in the commercial, industrial, and transport sectors. The forecast incorporates baseline and high-growth scenarios informed by national development frameworks and medium-term economic projections.
- iii. **Strategic Infrastructure and Industrialization Projects:** Major developments such as industrial parks as seen in Figure 3-1, mining operations, oil and gas infrastructure, and the Standard Gauge Railway (SGR) are key contributors to future demand. These projects are assumed to ramp up progressively, with their load impact factored in based on the expected year of commissioning and full operational capacity.
- iv. **Energy Transition and Electrification Programs:** Demand is also shaped by the shift toward e-mobility, electric cooking, and biomass substitution, particularly in public institutions and rural areas. These initiatives contribute to steady load growth and support clean energy access targets.

Together, these factors drive a diversified and rapidly evolving electricity demand landscape, requiring flexible planning to accommodate both organic growth and step-load additions.

In addition to baseline demand projections, the model integrates suppressed demand identified in the National Electrification Strategy (NES), which is estimated at approximately 2,420 MW nationwide. This includes unmet demand from unserved households, underserved public institutions (e.g., schools, health centres), and

unconnected industrial parks and SMEs.

The suppressed demand is spatially distributed based on NES electrification zones, with high concentrations observed in Karamoja, West Nile, Northern, and Eastern Uganda. These regions exhibit the highest deficits in access and infrastructure capacity. The model therefore integrates both actual and latent demand in each zone to ensure robust forecasting and infrastructure adequacy over time.

Demand projections also account for the expected activation of economic clusters outside the Greater Kampala Metropolitan Area (GKMP), where electricity consumption is projected to rise with the planned development of 25 industrial parks, oil and gas infrastructure, and region-specific mining and agro-processing activities.

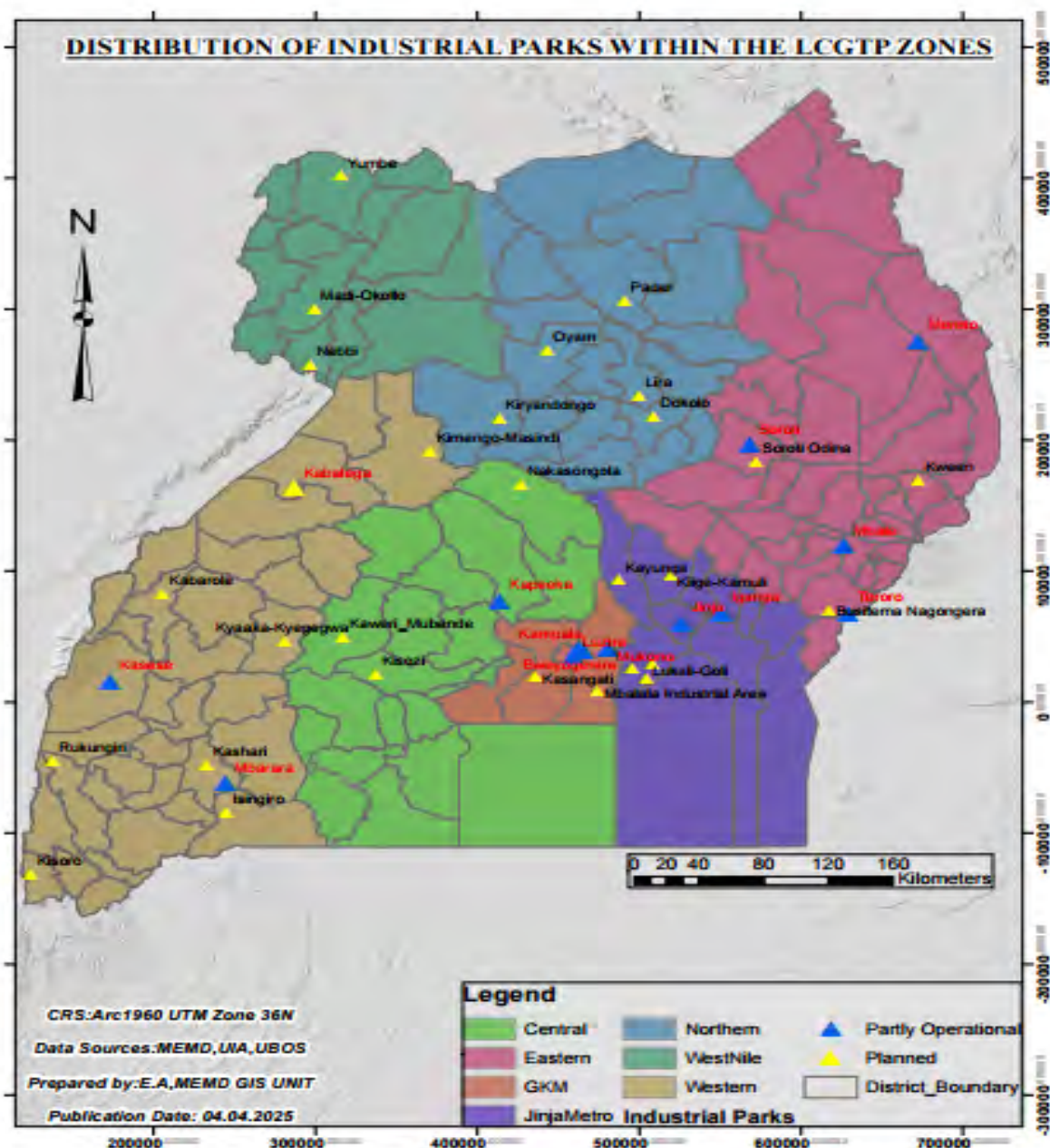


Figure 3-1: Distribution of industrial parks with the Uganda zones

The chart in Figure 3-2 below illustrates the key demand drivers under the 52 GW scenario from 2024 to 2040, highlighting the cumulative impact of emerging and traditional loads. While BAU demand and power exports grow steadily, the most significant contributors to future demand are biomass conversion, industrial parks, and e-mobility, especially beyond 2030. These strategic and policy-driven loads cause a sharp acceleration in total demand, with system requirements reaching approximately 35,000 MW by 2040, underscoring the importance of planning for large-scale infrastructure and energy transition initiatives.

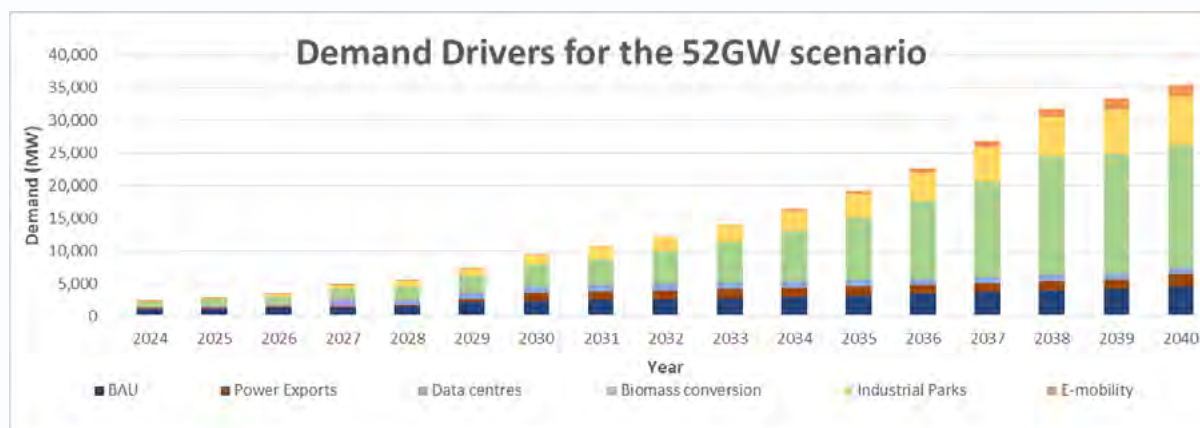


Figure 3-2: Major Demand Drivers

3.2.3 Distribution and Access Considerations in Demand Planning

The demand forecasts used in the LCGTP are shaped not only by macroeconomic and demographic factors but also by assumptions about the reach and functionality of the distribution network. In particular, the National Electrification Strategy (NES) identifies significant latent or suppressed demand in areas where households, businesses, and institutions lack physical access to electricity or experience unreliable supply.

To capture this, the LCGTP integrates the following considerations into its demand projections:

- The expansion and intensification of the distribution network, especially in low-access regions such as Karamoja, West Nile, and Teso, is assumed to unlock currently unmet demand and enable new service connections.
- The deployment of off-grid and mini-grid solutions is factored in for areas where grid extension is not immediately cost-effective. These interventions are guided by NES Tier classifications and electrification zone planning.
- The electrification of economic and social infrastructure including schools, health centres, water facilities, and trading centres is modeled in line with the Parish Development Model and government targets for universal access.
- These interventions are expected to stimulate local demand and support productive use in rural and peri-urban areas.

These assumptions ensure that the demand profile used in the LCGTP reflects the full spectrum of access-related interventions planned by the Government, and aligns with the multi-tier access and technology-neutral approach of the NES.

3.2.4 Interconnections with neighboring countries

Uganda's existing interconnections with neighboring countries provide important benefits, including system stability, grid support, and enhanced energy access in border regions. These interconnections also lay the groundwork for future regional power trade under the Eastern Africa Power Pool (EAPP) framework.

Currently, Uganda is interconnected with:

- i. Kenya via the 132 kV Bujagali-Tororo line, which supports mutual grid stability and energy exchange.
- ii. Rwanda through the Mbarara-Mirama-Shango 220 kV transmission line, enabling bilateral power transfers.
- iii. Tanzania via the 132 kV Masaka-Mwanza interconnection, primarily supporting electrification of border areas.

These existing lines are instrumental in serving cross-border communities, improving grid reliability, and facilitating regional cooperation in electricity access. They also contribute to managing demand variability and optimizing the utilization of Uganda's generation capacity, particularly hydropower.

3.2.5 Eastern Africa Power Pool

Uganda's participation in the Eastern Africa Power Pool (EAPP) is central to its long-term strategy to become a regional power exporter, enhance grid stability, and optimize resource sharing across borders. As part of this commitment, Uganda is implementing a series of cross-border transmission projects aimed at strengthening interconnection with neighboring countries, enabling electricity trade, and supporting regional energy security.

The planned projects include:

- i. A 220 kV interconnection with Kenya via the NELSAP corridor (Bujagali-Tororo and Mbarara-Mirama), with residual works ongoing and completion targeted by 2025/26.
- ii. Two 400 kV interconnections with the Democratic Republic of Congo (DRC): one via Nkenda-Mpondwe and another through Masaka-Kasese-Kasese Industrial Park, extending up to the DRC grid, with phased implementation through 2031/32.
- iii. A major 400 kV line to Tanzania through Masaka-Mutukula-Mwanza, supporting regional trade and reliability, expected by 2027/28.
- iv. A strategic 400 kV line to South Sudan via Olwiyo-Bibia-Juba, which includes significant substation investment, planned for commissioning by 2028/29.

These investments not only align with Uganda's export ambitions under the 52 GW scenario, but also support regional balancing, reliability, and long-term least-cost development under the EAPP framework. They form a critical part of the backbone for regional power pooling and will help unlock surplus generation capacity for trade.

3.3 FORECAST ASSUMPTIONS

The electricity demand forecast for the period 2024–2040 is based on a set of structured assumptions aligned with national development strategies, sector priorities, and spatial growth trends.

Key assumptions include:

1. Population Growth and Spatial Distribution

Based on UBOS projections and disaggregated at district level. Population growth is used to estimate biomass-to-electricity transition potential, with an assumed 25,000 MW of demand from biomass substitution (primarily electric cooking), closely tied to demographic trends.

2. Universal Access by 2030

The forecast assumes 100% electricity access (grid and off-grid combined) by 2030, in line with national targets. This drives significant growth in domestic demand, especially in currently underserved rural districts.

3. Economic Growth (GDP)

GDP growth assumed in the range of 5–6% annually, influencing commercial and industrial demand. Sector-specific growth rates are adjusted to reflect policy priorities and historic performance.

4. Step Loads and Strategic Infrastructure Projects

Large, discrete loads such as data centers are modeled separately due to their “step” nature. These are introduced progressively starting mid-2020s and are not evenly distributed year-on-year.

5. Tenfold Strategy (ATMs – Agriculture, Transport, Manufacturing)

The model incorporates significant demand growth from targeted investment in ATMs:

- Agriculture: mechanization and agro-processing clusters
- Transport: e-mobility (EVs, e-boda charging) and Standard Gauge Railway (SGR)
- Manufacturing: industrial parks, steel and cement plants

These sectors are treated as high-growth demand clusters under national policy.

6. Electrification of Cooking (Biomass Transition)

Electric cooking demand is tied to population density and access rate projections, contributing the largest new load segment in rural and peri-urban areas over time.

7. Power Exports

Regional exports grow from 170 MW (2024) to 2,500 MW (2040), based on planned interconnectors and EAPP integration.

8. Energy Efficiency and Demand Elasticity

Moderate improvements in appliance efficiency are assumed, offset by increasing household income and electrification of end-uses.

9. Hourly Load Profiles and Seasonal Variation

Representative days capture average and peak demand across seasons and customer categories.

10. Technology Adoption and Demand Shifts

Uptake of new technologies (e.g., induction cookers, EVs, C&I solar) and grid-defection trends are factored into long-term demand shaping.



3.4 LOSS LEVELS

Electricity losses are a key factor in generation and transmission planning, as they influence the amount of energy that must be produced to meet end-user demand.

In the base year, losses were at 17% for MV, and 4.8% for HV as seen in Table 3-2 below.

Table 3-2: Loss percentages

VOLTAGE LEVEL	BASE YEAR	2040
MV	17%	9%
HV	4.8%	3%

Aligned with the targets set in Uganda's Energy Policy 2023, the model assumes a gradual decline in losses toward: 9% for MV, and 3% for HV by the year 2040. These projected reductions are based on ongoing and planned investments in transmission line upgrades, reactive power compensation, substation modernization, and advanced grid operation practices. The assumed targets are consistent across all planning scenarios and reflect national ambitions to reduce system inefficiencies and improve power affordability.

3.5 RESULTS OF THE FORECAST

3.5.1 Forecasted electricity demand

As shown in the graph in Figure 3-3 and Table 3-3 below, Electricity consumption in Uganda is projected to grow significantly across all scenarios. Under the Business as Usual scenario, demand reaches 18,495 GWh by 2040. The 10-Fold Growth scenario increases demand to 101,642 GWh, driven by rapid industrialization and strategic infrastructure. The Energy Policy scenario reflecting full implementation of access, clean cooking, and electrified transport targets projects the highest demand at 191,788 GWh by 2040.

These results highlight the scale of growth the power system must be prepared to support under varying development pathways.

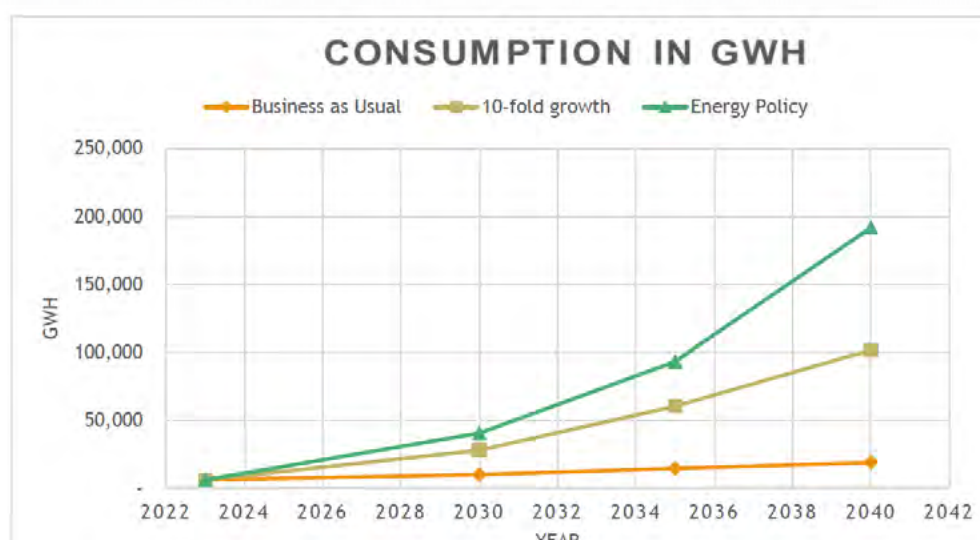


Figure 3-3: Forecasted electricity consumption for the different scenarios

Table 3-3: Forecasted electricity consumption in GWh

Scenario	Business as Usual	10-fold growth	Energy Policy
2023	5,732	5,732	5,732
2030	9,600	27,635	40,127
2035	14,073	60,160	93,053
2040	18,495	101,642	191,788

3.5.2 Peak demand

Figure 3-4 and Table 3-4 show Uganda's peak electricity demand rising from 882 MW in 2023 to 4,118 MW under Business as Usual, 17,138 MW under the 10-Fold Growth, and 38,189 MW under the Energy Policy scenario by 2040. Intermediate projections indicate demand reaching 1,864 –7,735 MW by 2030 and 2,932-19,843 MW by 2035, reflecting the scale of system growth required under different development pathways.

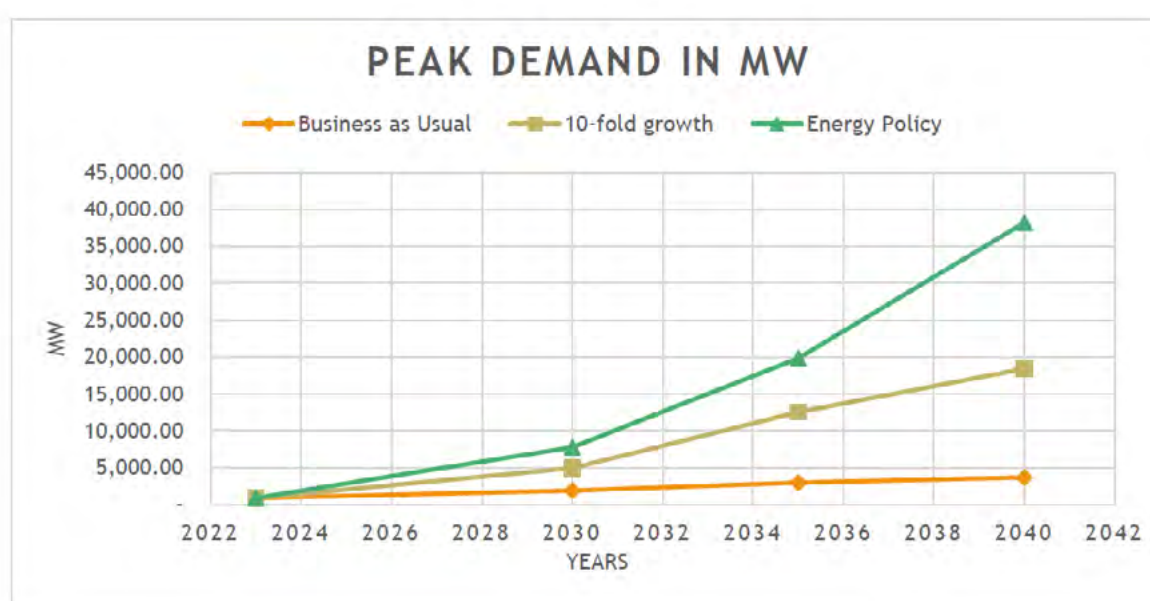


Figure 3-4: Peak demand projections over the years

Table 3-4: Electricity demand projection in MW 2025 - 2040

Scenario	Business as Usual	10-fold growth	Energy Policy
2023	882	882	882
2030	1,864	4,918	7,735
2035	2,932	12,517	19,843
2040	3,667	18,405	38,189

These projections reflect varying assumptions in industrialization, access expansion,

and electrification of end-uses. The Energy Policy scenario shows the steepest growth trajectory, underscoring the infrastructure requirements for full implementation of Uganda's energy transition targets.

3.5.3 Per Capita consumption

Table 3-5: Per Capita consumption consideration.

Scenario	Year	2023	2030	2035	2040
BAU	Peak demand (MW)	882	1,864	2,932	3,667
	Energy (GWh)	5,732	9,600	14,073	18,495
	Population	44,200,000	53,992,059	62,288,341	71,859,409
	Per capita (kWh)	130 ²	178	226	257
10-Fold	Peak demand (MW)	882	4,918	12,517	18,405
	Energy (GWh)	5,732	27,635	60,160	101,642
	Population	44,200,000	53,992,059	62,288,341	71,859,409
	Per capita (kWh)	130	512	966	1,414
Energy Policy	Peak demand (MW)	882	7,735	19,843	38,189
	Energy (GWh)	5,732	40,127	93,053	191,788
	Population	44,200,000	53,992,059	62,288,341	71,859,409
	Per capita (kWh)	130	743	1,494	2,669

As seen in Table 3-5 above, per capita consumption by 2040 reaches 257 kWh under Business as Usual, 1,414 kWh under the 10-Fold Growth scenario, and 2,669kWh under the Energy Policy scenario. These per capita values, applied to Uganda's growing population, form the basis for estimating total system energy and peak demand, reflecting the influence of access, economic activity, and electrification trends across scenarios.

A detailed table showing the contribution of each demand driver to per capita electricity consumption over the planning horizon (2024–2040) is provided in Appendix 14.2 of this report.

3.5.4 Conclusion for the demand forecast

Uganda's electricity demand is projected to grow significantly over the planning horizon, driven by rising population, urbanization, electrification, industrialization, and energy access targets under the Energy Policy 2023. However, actual demand

growth may vary due to shifting consumption patterns, including the uptake of captive generation, direct power purchase arrangements, and net metering, especially among large and commercial users.

The increasing deployment of self-generation systems, particularly solar PV without storage, is leading to reduced reliance on grid power during daylight hours. This trend, while improving energy access and affordability for end-users, poses risks to grid-based sales and may result in deemed energy where available generation cannot be dispatched due to insufficient demand or infrastructure limitations. Additionally, policy reforms that promote Independent Power Transmission Operators (IPTOs) and open-access wheeling could further decentralize demand and shift it outside the traditional utility framework.

To ensure sustainable and predictable demand growth that supports long-term system investments, the following recommendations are made:

- i. Fast-track the implementation of industrial parks, oil and gas infrastructure, and other strategic projects that anchor large-scale, stable demand.
- ii. Expand demand creation initiatives, especially in the productive sectors of agriculture, transport, and manufacturing, under the Tenfold Strategy.
- iii. Develop frameworks for integrating emerging models, such as direct sales, net metering, and distributed generation, without undermining utility revenue stability.
- iv. Ensure accurate demand forecasting and infrastructure planning to avoid stranded capacity and mitigate the financial impacts of deemed energy.
- v. Promote grid reliability and tariff competitiveness to retain large consumers on the grid while complementing decentralized solutions.
- vi. Enhance coordination between sector institutions and local governments to align electrification efforts with economic activity and load growth.
- vii. These measures are necessary to balance flexibility and grid stability, while aligning demand-side growth with Uganda's broader development and energy transition goals.

3.6 ELECTRICITY GENERATION SOURCES

This chapter presents the energy sources planned for Uganda's future electricity generation. It covers both conventional and emerging technologies such as hydropower, solar, wind, thermal, biomass, and nuclear selected based on cost, availability, reliability, and alignment with national policy goals.

3.6.1 Least-Cost Generation Technology Screening

The selection of generation technologies in this Plan was guided by a least-cost optimization approach, ensuring that future investments deliver reliable power at the lowest possible system cost, while aligning with Uganda's energy policy and climate goals. Technologies were screened and ranked based on a set of economic, technical, and environmental criteria, with only viable and policy-aligned options retained for detailed modeling.

A. Fuel Prices

Table 3-6 below provides the fuel prices for various electricity generation technologies considered in the LCGTP. These prices are critical inputs for the Electricity Planning Model, as they influence the cost of electricity generation and the optimization of the energy mix. The prices reflect the cost of fuel required to operate different types of power plants.

Table 3-6: Fuel prices for the different technologies

Fuel prices (\$/MMBTU)	
Nuclear	0.878
Natural gas	10
Heavy Fuel Oil	25.09
Diesel	40.29
Biomass	5

Source: Ministry of Energy and Mineral Development

B. Capital Expenditure (CAPEX)

CAPEX values for the different technologies are input in the “CapexTrajectories” sheet. These figures provide the estimated capital costs (in USD/kW or USD/MW) for each generation technology across key planning years and the fixed operation and maintenance costs. These values inform technology selection in the least-cost model by reflecting upfront investment needs.

C. Operational Expenditure (OPEX)

OPEX data comprising both the fuel cost and variable operation and maintenance costs is found in the “Generator Data” sheet. This includes plant-specific technical and economic parameters, such as heat rates, fuel type, etc. OPEX figures help determine long-term operating costs within the dispatch optimization.

D. Availability Factor / Capacity Factor

The availability of generators is defined on a monthly (seasonal) basis depending on technology and resource availability. This allows the model to account for seasonal fluctuations such as hydropower output during dry months and reduced availability due to maintenance.

3.6.2 RESOURCE ASSESSMENT

As stipulated in the Energy policy 2023, the Table 3-7 below outlines Uganda’s projected resource potential by 2040, distributed across various energy technologies. It shows the planned generation mix with nuclear (45.7%), solar (19%), and hydro (8.6%) as the leading sources. It reflects a diversified strategy that incorporates both conventional and emerging technologies, including green hydrogen, waste-to-energy, and BESS for system flexibility.

Table 3-7: Total capacity per technology

Resource	Capacity (MW)	Percentage
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Hydro	4,500	8.6%
Geothermal	1,500	2.9%
Solar	10,000	19.0%
Nuclear	24,000	45.7%
Co-Generation	1,000	1.9%
Green Hydrogen	5,000	9.5%
Gas	3,000	5.7%
Waste to Energy	1,000	1.9%
Wind	200	0.4%
Thermal	1,500	2.9%
Peat	800	1.5%
BESS	Infinite	
Total capacity	52,500	

Source: Energy Policy 2023

3.6.3 Committed Generation Plants

In the context of this Plan, committed projects refer to electricity generation projects that are at an advanced stage of development and are considered certain to be completed and commissioned within the short to medium term.

The Table 3-8 presents the committed projects.

Table 3-8: Committed projects

No.	Project Description	Region	Capacity (MW)	COD
	Construction of a 3.3 MW small hydropower plant at Hoimo	West	3.3	2027
	Construction of a 0.3 MW small hydropower plant at Igassa	West	0.3	2027
	Construction of a 0.7 MW small hydropower plant at Nsongya	West	0.7	2027
	Construction of a 3.2 MW small hydropower plant at Nyabuhuka	West	3.2	2026
	Construction of a 7.0 MW small hydropower plant at Nyamabuye	West	7.0	2028
	Construction of a 0.3 MW small hydropower plant at Tokwe	West	0.3	2027

	Construction of a 0.4 MW small hydropower plant at Kabasanja	Central	0.4	2027
	Construction of a 0.3 MW small hydropower plant at Katooke	Central	0.3	2027
	Construction of a 6.5 MW small hydropower plant at Kabeywa I	East	6.5	2028
	Construction of a 2.0 MW small hydropower plant at Kabeywa II	East	2.0	2029
	Construction of a 0.5 MW small hydropower plant at Nchwera	West	0.5	2027
	Construction of a 9.5 MW small hydropower plant at Simu	East	9.5	2027
	Construction of a 7.0 MW small hydropower plant at Sisi East		7.0	2027
	Construction of a 0.5 MW small hydropower plant at Warugo	West	0.5	2028
	Run-of-River Subtotal		51.3	
	Construction of a 20.0 MW wind power plant at Rupa	East	20.0	2025
	Wind Subtotal		20.0	
	Construction of a 9.5 MW bagasse cogeneration plant at SCOUL (Jinja Metro)	Central	9.5	2025
	Biomass Subtotal		9.5	
	Construction of a 20.0 MW solar PV plant at Ituka	West Nile	20.0	2026
	Solar PV Subtotal		20.0	

3.6.4 Candidate Generation Projects

Candidate plants are long-term prospects that may not yet have completed feasibility studies or secured financing. Their development is subject to system needs, cost optimization, and policy priorities. These plants are evaluated by the model for potential inclusion in least-cost scenarios.

Table 3-9 presents Uganda's pipeline of candidate plants identified for potential long-term development. The projects are categorized by resource type, with indicative capacities

Table 3-9: Candidate Generation Projects

No.	Project Description	Region	Capacity (MW)	COD	Current Status	Financier
	Construction of an 840 MW hydropower plant at Ayago	West Nile	840	2032	Feasibility studies are scheduled to start in the FY 25/26	GOU under the PIMs Plus Project
	Construction of a 392 MW hydropower plant at Oriang	North	392	2033	Feasibility studies are scheduled to start in the FY 25/26	GOU under the PIMs Plus Project
	Construction of a 400 MW hydropower plant at Kiba	West Nile	400	2031	Feasibility studies are scheduled to start in the FY 25/26	GOU
	Construction of a 390 MW hydropower plant at Uhuru	North	390	2036	Feasibility studies yet to be conducted	To be determined
	Construction of a 52 MW small hydropower plant at Achwa-Aber	North	52	2029	Feasibility studies yet to be conducted	To be determined
	Construction of a 10.5 MW small hydropower plant at Agago-Achwa IV	North	10.5	2029	Feasibility studies yet to be conducted	To be determined
	Construction of a 3.3 MW small hydropower plant at Atari	East	3.3	2028	Feasibility studies yet to be conducted	To be determined
	Construction of an 18 MW small hydropower plant at Awere	North	18	2029	Feasibility studies yet to be conducted	To be determined
	Construction of a 7 MW small hydropower plant at Buwangani	East	7	2028	Feasibility studies yet to be conducted	To be determined
	Construction of a 5 MW small hydropower plant at Kiraboha	West	5	2027	Feasibility studies yet to be conducted	To be determined

	Construction of a 2.5 MW small hydropower plant at Kisinga	West	2.5	2028	Feasibility studies yet to be conducted	To be determined
	Construction of a 4.2 MW small hydropower plant at Latoro, Nwoya district	West Nile	4.2	2030	The Ministry received & processed the implementation agreement	To be determined
	Construction of a 17.4 MW small hydropower plant at Lower Achwa	North	17.4	2029	Feasibility studies yet to be conducted	To be determined
	Construction of a 6.7 MW small hydropower plant at Lwakhakha	East	6.7	2030	Feasibility studies yet to be conducted	To be determined
	Construction of a 1.2 MW small hydropower plant at Maziba	West	1.2	2027	Feasibility studies yet to be conducted	To be determined
	Construction of a 19 MW cluster of small hydropower plants (Mini Hydros 3)	East	19	2031	Feasibility studies yet to be conducted	To be determined
	Construction of a 13.6 MW small hydropower plant at Mitano	West	13.6	2027	Feasibility studies yet to be conducted	To be determined
	Construction of a 3.4 MW small hydropower plant at Mukoki	West	3.4	2028	Feasibility studies yet to be conducted	To be determined
	Construction of a 50 MW hydropower plant at Muzizi	West	50	2030	PPP requested by (MEMD, UEGCL)	GoU
	Construction of a 7.5 MW small hydropower plant at Nengo Bridge	West	7.5	2027	The Ministry received and processed the implementation agreement	To be determined
	Construction of a 13.8 MW small hydropower plant at Ngenge	West	13.8	2030	Feasibility studies yet to be conducted	To be determined

Construction of a 39 MW small hydropower plant at Nshongezi	West	39	2027	Feasibility studies scheduled to start in FY 2026/27	AFDB
Construction of a 7 MW small hydropower plant at Nsongi	West	7	2027	Feasibility studies yet to be conducted	To be determined
Construction of a 5.2 MW small hydropower plant at Nyakinengo	West	5.2	2028	Feasibility studies yet to be conducted	To be determined
Construction of a 5 MW small hydropower plant at Okollo	West Nile	5	2029	The ministry received and is processing implementation agreements for the plant	To be determined
Construction of a 0.4 MW small hydropower plant at Rwembya	West	0.4	2030	Feasibility studies yet to be conducted	To be determined
Construction of a 7 MW small hydropower plant at Sironko	East	7	2027	Feasibility studies yet to be conducted	To be determined
Construction of thirty (30) small hydropower plants in Western Uganda	West	180	2029	Feasibility studies yet to be conducted	To be determined
Construction of a 580 MW solar PV plant in Central Uganda	Central	580	2030	Feasibility studies yet to be conducted	To be determined
Construction of a 20MW solar PV plant in Central Uganda, Kapeka	Central	20	2030	Permits have been received from ERA to conduct feasibility Studies	Total Energies
Construction of a 5,550 MW solar PV plant in Eastern Uganda	East	5,550	2030	Feasibility studies yet to be conducted	To be determined

	Construction of a 20MW solar PV plant in Eastern Uganda, Pallisa district	East	20	203	Permits have been received from ERA to conduct feasibility Studies	Total Energies
	Construction of a 20MW solar PV plant in Eastern Uganda, Bukedea district	East	20	20	Permits have been received from ERA to conduct feasibility Studies	Total energies
	Construction of a 20MW solar PV plant in Eastern Uganda, Tororo district	East	20	2030	Permits have been received from ERA to conduct feasibility Studies	Total Energies
	Construction of a 20MW solar PV plant in Eastern Uganda, Iganga District	East	20	2030	Permits have been received from ERA to conduct feasibility Studies	Total Energies
	Construction of a 20MW solar PV plant in Eastern Uganda, Kumi District	East	20	2030	Permits have been received from ERA to conduct feasibility Studies	Total Energies
	Masdar Solar Power project 150MW grid-tie plant in Bulambuli district	East	150	2030	Feasibility studies yet to be completed	Masdar
	4MW solar power plant installed at Busitema University	East	4	2030	Ongoing	GoU

	Photon Power Limited solar power plant, phased 100MW, starting with 20MW in the Soroti district	East	100	2030	•Policy directive has been gazetted •Photon Power is in the process of applying for a feasibility study permit	Triba Energy
	Construction of a 1,150 MW solar PV plant in Northern Uganda	North	1,150	2030	Feasibility studies yet to be conducted	To be determined
	Machame Energy Limited 20MW solar power plant in Nwoya district	North	20	2030	A feasibility permit has been issued by ERA, and a full feasibility study is ongoing.	Machame Energy and GoU
	Karamoja Solar Company Uganda Limited (KSCL) solar power plant, 50MW, Moroto district.	North	50		•A pre-feasibility study was carried out and submitted to MEMD •A policy directive was given by the Minister for ERA to issue a feasibility study permit to begin studies in December 2024.	KSCL (with an investor)
	Construction of a 1,610 MW solar PV plant in West Nile	West Nile	1,610	2030	Feasibility studies yet to be conducted	To be determined
	Construction of a 417 MW solar PV plant in Western Uganda	West	417	2030	Feasibility studies yet to be conducted	To be determined

	Kasese Cobalt Company Limited (KCCL) 20MW solar power plant in Kasese district	West	20	2030	•Pre-feasibility study submitted to MEMD •A policy directive was given by the Minister for ERA to issue a feasibility permit in November 2024	KCCL
	Construction of a 250 MW floating solar PV plant at Jinja	Jinja Metro	250	2030	Feasibility studies yet to be conducted	To be determined
	Construction of a 10 MW Floating Solar pilot plant at Isimba Dam, Kamuli District	East	10	2030	Feasibility studies are going on	GoU (SwedFund)
	Construction of the Vindulaka 10MW floating solar on Lake Victoria, Entebbe	Central	10		Land was acquired, and a pre-feasibility study and consultations ongoing.	To be determined
	Construction of a 20 MW wind power plant in Napak	East	20	2026	Feasibility studies completed	Senok
	Construction of a 120 MW wind power plant in Karamoja	East	120	2036	Feasibility studies yet to be conducted	To be determined
	Construction of an 80 MW wind power plant in West Nile	West Nile	80	2026	Feasibility studies yet to be conducted	To be determined

	Construction of a 7,200 MW nuclear power plant at Nakasongola	Central	7,200	2036	Feasibility studies yet to be conducted	To be determined
	Construction of an 8,400 MW nuclear power plant at Buyende	East	8,400	2032	Feasibility studies for FY 2025/26	To be determined
	Construction of a 4,000 MW nuclear power plant at Lamor	North	4,000	2046	Feasibility studies yet to be conducted	To be determined
	Construction of a 4,800 MW nuclear power plant at Kiruhura	West	4,800	2041	Feasibility studies yet to be conducted	To be determined
	Construction of a 20 MW biogas power plant at Masindi	West	20	2028	Feasibility studies yet to be conducted	To be determined
	Construction of a 20 MW biomass power plant at Pece	North	20	2028	Feasibility studies yet to be conducted	To be determined
	Construction of a 10 MW biomass power plant at Victoria Sugar	—	10	2028	Feasibility studies yet to be conducted	To be determined
	Construction of a 9 MW bagasse cogeneration plant at Kyankwanzi	West	9	2028	Feasibility studies yet to be conducted	To be determined
	Construction of a 20 MW biomass power plant at Unergy Bio	Central	20	2028	Feasibility studies yet to be conducted	To be determined
	Construction of a 200 MW biomass power plant at Central Bio	Central	200	2026	Feasibility studies yet to be conducted	To be determined

	Construction of a 2.5 MW waste-to-energy plant at Jinja	East	2.5	2026	Feasibility studies yet to be conducted	To be determined
	Construction of a 3 MW bagasse cogeneration plant at Kamuli Sugar	East	3	2027	Feasibility studies yet to be conducted	To be determined
	Construction of an 8 MW bagasse cogeneration plant at Pramukh	East	8	2026	Feasibility studies yet to be conducted	To be determined
	Construction of a 20 MW biomass power plant at Pece	North	20	2027	Feasibility studies yet to be conducted	To be determined
	Construction of a 30 MW peat-fired power plant at Kabale	West	30	2029	Feasibility studies yet to be conducted	To be determined
	Construction of a 30 MW bagasse cogeneration plant at Kinyara Sugar	East	30	2027	Received Solicitor General's Clearance	To be determined
	Construction of a 50 MW peat-fired power plant at Kyoga	West	50	2028	Feasibility studies yet to be conducted	To be determined
	Construction of a 620 MW biomass complex in Western Uganda	West	620	2028	Feasibility studies yet to be conducted	To be determined
	Construction of a 461 MW biomass complex in Eastern Uganda	East	461	2028	Feasibility studies yet to be conducted	To be determined
	Construction of a 150 MW geothermal power plant at Panyimur	West Nile	150	2027	Feasibility studies yet to be conducted	To be determined

Construction of a 680 MW geothermal power plant at Katwe	West	680	2030	Feasibility studies yet to be conducted	To be determined
Construction of a 250 MW geothermal power plant at Kibiro	West	250	2028	Feasibility studies yet to be conducted	To be determined
Construction of a 90 MW geothermal power plant at Elegu	North	90	—	Feasibility studies yet to be conducted	To be determined
Construction of a 380 MW geothermal power plant at Buranga	West	380	2028	Feasibility studies yet to be conducted	To be determined
Construction of a 1,250 MW combined cycle gas turbine (CCGT) power plant in Eastern Uganda	East	1,250	2030	Feasibility studies yet to be conducted	To be determined
Construction of a 1,500 MW combined cycle gas turbine (CCGT) power plant in Western Uganda	West	1,500	2030	Feasibility studies yet to be conducted	To be determined
Construction of a 1,250 MW open cycle gas turbine (OCGT) power plant in Eastern Uganda	East	1,250	2030	Feasibility studies yet to be conducted	To be determined
Construction of a 1,646 MW open cycle gas turbine (OCGT) power plant in Western Uganda	West	1,646	2030	Feasibility studies yet to be conducted	To be determined
Construction of grid-scale battery energy storage systems (BESS)	National	1,551	2030	Feasibility studies yet to be conducted	To be determined



4 PLANNING ANALYSIS

4.1 PLANNING SCENARIOS

The table 4-1 outlines the three energy demand scenarios used in this Plan to guide generation and transmission expansion through 2040.

The BAU scenario targets 3,667 MW of installed capacity by 2040, reflecting moderate demand growth and reliance on existing infrastructure. The Tenfold Strategy projects 18,405 MW, aligned with accelerated industrialization. The Energy Policy 2023 scenario is the most ambitious, targeting 38,189 MW, driven by biomass transition, large-scale infrastructure, and digital economy growth.

Table 41: Energy Demand Scenarios Used in the Least-Cost Power Development Plan

Scenario	Nomi- nal GDP Growth Rates	Demand growth rate	Peak demand by 2040	Assumptions	Remarks
Business As Usual	6.5%	10.0%	3,667 MW	Optimization of the existing infrastructure in key industrial parks and implementation of the electricity connections policy	Based on historical data. Demand constrained by current pace of ESI infrastructure development.
Tenfold Growth strategy	14%	15.3%	18,405 MW	ESI development trajectory with an electricity demand growth rate following the Government's Strategy to grow the economy ten-fold	Optimization of Govern-ment's indus-trialization and social transforma-tion agenda.
Energy poli-cy 2023	17%	18.4%	38,189 MW	ESI development trajectory based on the Energy Policy 2023.	Transition from biomass (87.8% of the energy mix), Data centers, EACOP, Mining, Industrial Parks and so on etc.

4.2 GENERATION EXPANSION PLANNING

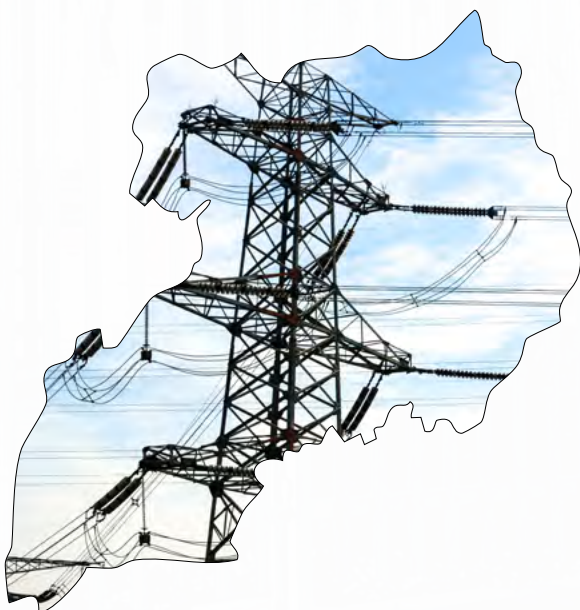
The generation expansion planning in this Plan is carried out using EPM. The model selects and schedules generation investments to meet hourly electricity demand over the planning horizon, subject to technical, economic, and policy constraints.

EPM considers projected demand from all scenarios, system reserve requirements, capacity limits, fuel prices, emissions, and technology-specific characteristics such as CAPEX, OPEX, availability, and dispatchability. It evaluates the timing, type, and scale of generation projects needed to ensure reliability at the lowest possible cost, while also incorporating grid constraints, ancillary service needs, and carbon pricing. The model co-optimizes generation with transmission decisions to ensure system adequacy and regional balancing across zones.

4.3 TRANSMISSION PLANNING

Transmission planning in this Plan is carried out alongside generation planning using a multi-zonal power system model. The model optimizes power flows between these zones, taking into account existing and planned transmission infrastructure, including thermal limits, transformer capacities, and inter-zonal transfer capabilities. Transmission constraints and bottlenecks are identified based on the ability of each corridor to evacuate power from generation sources to load centers under both normal and contingency (N-1) conditions.

Where inter-zonal transfer capacity is insufficient to meet least-cost generation dispatch, the model recommends reinforcements or new lines to reduce congestion, improve reliability, and facilitate power trade. This co-optimization of generation and transmission ensures that the system can efficiently balance supply and demand across the country, while minimizing investment costs and maintaining grid stability.



This co-optimization of generation and transmission ensures that the system can efficiently balance supply and demand across the country, while minimizing investment costs and maintaining grid stability.





5 RESULTS

5.1 GENERATION MIX FOR DIFFERENT SCENARIOS

5.1.1 Generation mix in GWh

The Table 5-1 below presents Uganda's projected electricity generation mix under the different planning scenarios. It details the major electricity generation technologies and their respective contributions in gigawatt-hours (GWh) and percentage share of the total generation.

Table 5-1: 2040 Generation Energy mix for the different scenarios

SCENARIOS		GENERATION MIX		MAJOR ELECTRICITY 2040 GENERATION TECHNOLOGY	
BAU - electricity demand grows averagely at 10%		Hydro	15,999 GWh (81.8%)	Nuclear	43,355 GWh (41.0%)
		CCGT	1,024 GWh (5.2%)	Hydro	27,410 GWh (25.9%)
		PV	945 GWh (4.8%)	PV	11,506 GWh (10.9%)
		Biomass	859 GWh (4.4%)	Geothermal	9,509 GWh (9.0%)
		Others (HFO, Wind, etc)	745 GWh (3.8%)	Others	13,660 GWh (13.1%)
10-fold growth - demand grows at 15%		Nuclear	43,355 GWh (41.0%)	Nuclear	104,519 GWh (51.8%)
		Hydro	27,410 GWh (25.9%)	Hydro	27,410 GWh (13.6%)
		PV	11,506 GWh (10.9%)	CCGT	14,331 GWh (7.1 %)
		Geothermal	9,509 GWh (9.0%)	Solar	13,396 GWh (6.6%)
		Others	13,660 GWh (13.1%)	Others (OCGT, Geo- thermal, BESS, Wind & HFO)	42,099 GWh (20.9%)
Energy Policy 2023		Nuclear	104,519 GWh (51.8%)		
		Hydro	27,410 GWh (13.6%)		
		CCGT	14,331 GWh (7.1 %)		
		Solar	13,396 GWh (6.6%)		
		Others (OCGT, Geo- thermal, BESS, Wind & HFO)	42,099 GWh (20.9%)		
Nuclear Wind PV HFO Diesel BESS Geothermal Biomass CCGT OCGT RoR					

The table presents projected generation mix under each of the three demand scenarios Business as Usual (BAU), 10-Fold Growth, and the Energy Policy 2023 case;

- In the BAU scenario, electricity supply is dominated by hydropower (81.8%), with limited contributions from thermal, solar, and biomass.
- Under the 10-Fold Growth scenario, nuclear energy (41.0%) emerges as a major base load source, while hydropower, solar, and geothermal also scale up significantly.
- In the Energy Policy 2023 scenario, nuclear generation accounts for over half (51.8%) of electricity production, supported by a diversified mix including hydro (13.6%), CCGT (7.1%), and solar (6.6%), with the remainder supplied by other flexible and renewable technologies.

These mixes reflect a strategic shift toward cleaner, large-scale, and dispatchable energy sources to meet rising demand and policy objectives.

5.1.2 Generation capacity mix in GW

Table 5-2 provides a comparative overview of the projected installed capacity by 2040 across these scenarios, highlighting key differences in technology adoption, renewable energy penetration, and reliance on fossil fuels.

Table 5-2: 2040 Generation capacity mix for the different scenarios

Element	BAU	10 Fold	Energy Policy
Total Generation Capacity (MW)	4,602.6	27,125.4	41,348.1
Top 3 Contributors (% in 2040)	Hydro (63.7%) PV (13.5%) OCGT (11.2%)	PV (27.0%) Nuclear (19.6%) Hydro (17.2%)	Nuclear (31.3%) PV (21.0%) Hydro (11.5%)
Nuclear	x	(5,030.0 MW)	(12,000 MW)
Geothermal	x	(1,210 MW) - 4.7%	(1,210 MW) - 2.85%
Fossil Fuels ³ Share - 2040	~18.4%	~30.0%*	~28.6%*
Renewables Share (2040)	~81.6%	~70.0%	~71.4%

³Fossil Fuels = CCGT, OCGT, Diesel & HFO



As seen in the Table above, the Energy Policy scenario shows the most ambitious shift towards low-carbon technologies, driven by significant nuclear and solar PV investments and reduced fossil fuel dependence. In contrast, the BAU scenario maintains a relatively conservative trajectory, heavily reliant on hydro and peaking plants like PV & OCGT. The 10-Fold scenario balances aggressive capacity expansion with a diversified mix that includes PV, nuclear, and hydro, while still maintaining a notable fossil fuel share.

5.2 TRANSMISSION NETWORK GROWTH FOR THE DIFFERENT SCENARIOS

This section presents the projected cumulative and inter-zonal transmission capacity requirements under each scenario.

5.2.1 Transmission Capacity Expansion

Table 5-3 presents the cumulative national transmission capacity needed to meet projected electricity demand across the three scenarios. Under the Business as Usual (BAU) case, transmission capacity reaches 12,226 MW by 2035 and remains stable thereafter. The Tenfold Growth and Energy Policy 2023 scenarios require substantial scaling, with national transmission capacity growing to 30,946 MW and 44,886 MW, respectively, by 2040. This reflects the scale of infrastructure required to reliably evacuate and deliver energy from generation zones to load centers.

Table 5-3: Projected Cumulative Transmission Capacity (MW) under the different growth scenarios

SCE- NARIO	CUMULATIVE TRASMISSION CAPACITY (MW)		
	2030	2035	2040
Business as Usual	7,937	12,226	12,226
10-fold growth	10,277	23,926	30,946
Energy Policy	22,937	34,886	44,886

5.2.2 Inter-Zonal Transfer Requirements

To ensure a balanced and reliable system, the Plan also evaluates the inter-zonal transmission capacities between Uganda's seven planning zones. Table 5-4 provides detailed projections of these capacities for 2030, 2035, and 2040 across all scenarios.

Table 5-4: Projected inter zonal transmission capacities between the different zones

ZONE	SCENARIO	2023	2030	2035	2040
CENTRAL - GKM	BAU	497	497	497	497
	10-FOLD	497	497	5117	9857
	ENERGY POLICY	497	5497	5497	10497
CENTRAL - WEST	BAU	192	884	3224	3224
	10-FOLD	192	3224	5564	5564
	ENERGY POLICY	192	5884	5884	5884
EAST-NORTH	BAU	117	2574	2574	2574
	10-FOLD	117	2574	2574	2574
	ENERGY POLICY	117	2574	2574	7574
GKM-NORTH	BAU	2340	2340	2340	2340
	10-FOLD	2340	2340	4680	7020
	ENERGY POLICY	2340	7340	7340	7340
JINJA METRO- EAST	BAU	140	482	482	482
	10-FOLD	140	482	2822	2822
	ENERGY POLICY	140	482	5482	5482
JINJA METRO- GKM	BAU	480	480	480	480
	10-FOLD	480	480	480	480
	ENERGY POLICY	480	480	5480	5480
WEST NILE - NORTH	BAU	679	679	2628	2628
	10-FOLD	679	679	2628	2628
	ENERGY POLICY	679	679	2628	2628

Under BAU, most corridors remain unchanged beyond 2030. In contrast, the Tenfold and Energy Policy cases require major upgrades especially along GKM–North, Jinja Metro–East, and Central–GKM routes to accommodate increased power flows and provide system redundancy.

These reinforcements are critical for enabling least-cost generation dispatch, supporting regional power balance, and maintaining grid stability as Uganda transitions to a high-growth and diversified energy future.

Under each scenario, the model ensures that electricity supply reliably meets projected demand throughout the planning horizon. This is achieved by enforcing a strict supply-demand balance constraint, which requires that total generation (including imports and reserves) is sufficient to meet hourly demand across all zones. Any shortfall is captured as unmet demand, which the model minimizes through least-cost dispatch and timely generation expansion. This mechanism ensures system adequacy while optimizing investment in new capacity, accounting for variability, transmission limits, and operational constraints.

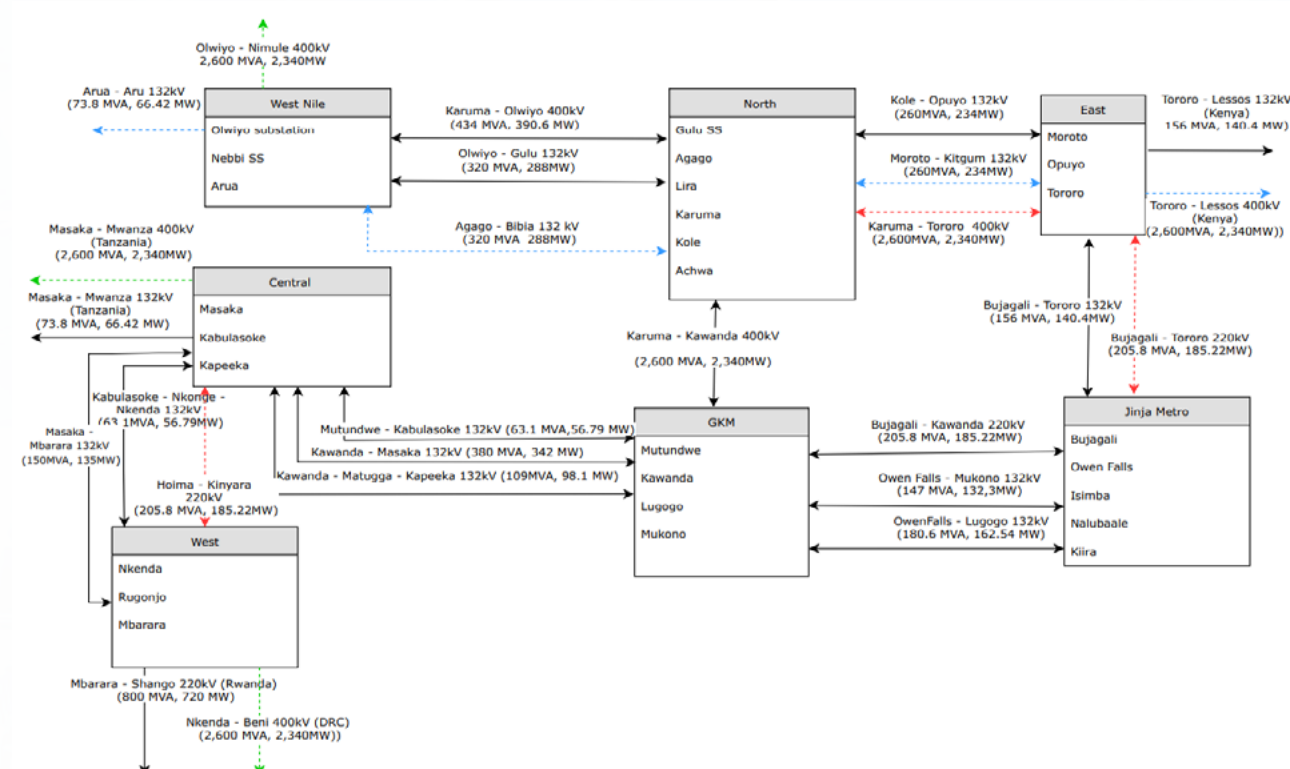
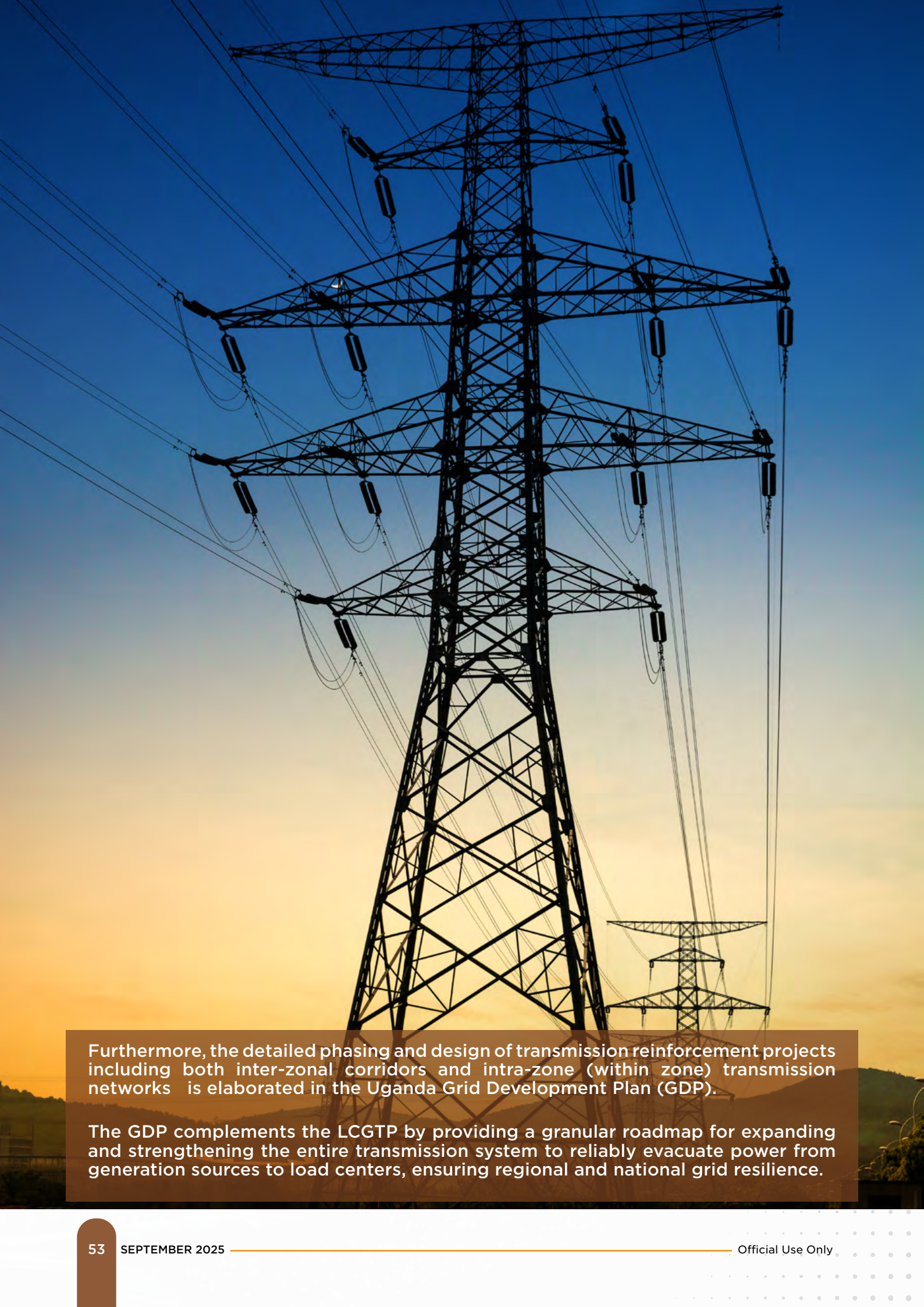


Figure 5-1: Planned Interzonal Power Transfer TX Lines_BAU

From the figure 5-1 above, the red lines represent highly critical transmission infrastructure essential for power evacuation from Karuma Hydropower Plant and regional interconnection with Kenya. The Karuma-Tororo 400kV line facilitates bulk power transfer to Eastern Uganda and connects to the Tororo-Lessos 400kV line, enabling electricity trade through the Eastern Africa Power Pool (EAPP). These lines handle large capacities (up to 2,340 MW), making them pivotal for national grid stability and regional export. On the other hand, the blue lines like Karuma-Oliwyo 400kV and Agago-Bibia 132kV are semi-critical they support supply to the West Nile and Northern Uganda, and future integration with South Sudan, but serve mainly to reinforce localized reliability rather than national or cross-border energy flows.



Furthermore, the detailed phasing and design of transmission reinforcement projects including both inter-zonal corridors and intra-zone (within zone) transmission networks is elaborated in the Uganda Grid Development Plan (GDP).

The GDP complements the LCGTP by providing a granular roadmap for expanding and strengthening the entire transmission system to reliably evacuate power from generation sources to load centers, ensuring regional and national grid resilience.

5.3 FLEXIBILITY OF THE SYSTEM

The dispatch charts for the year 2040 across the three scenarios BAU, Tenfold Strategy, and Energy Policy 2023 in Figure 5-2, 5-3 and 5-4 respectively highlight the role and flexibility of various generation sources in meeting hourly demand.

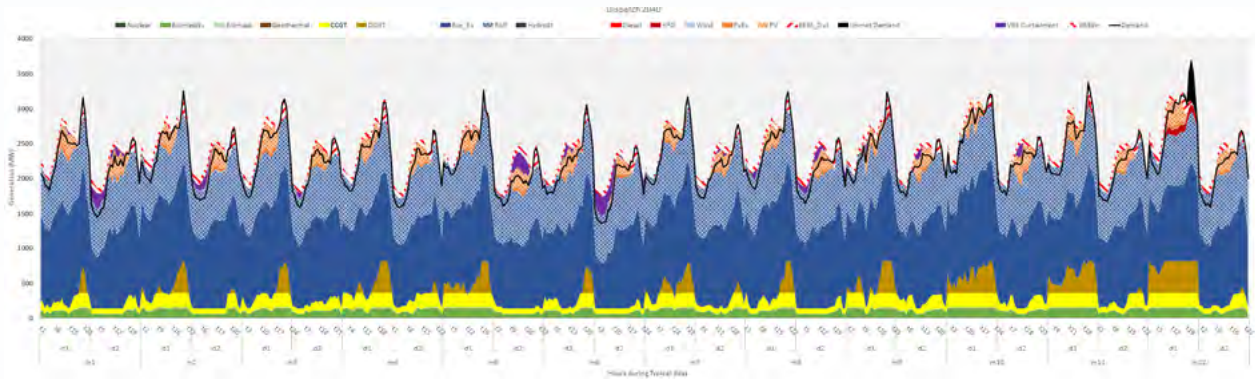


Figure 5-2: 2040 Hourly Dispatch over the representative days each month for the BAU scenario

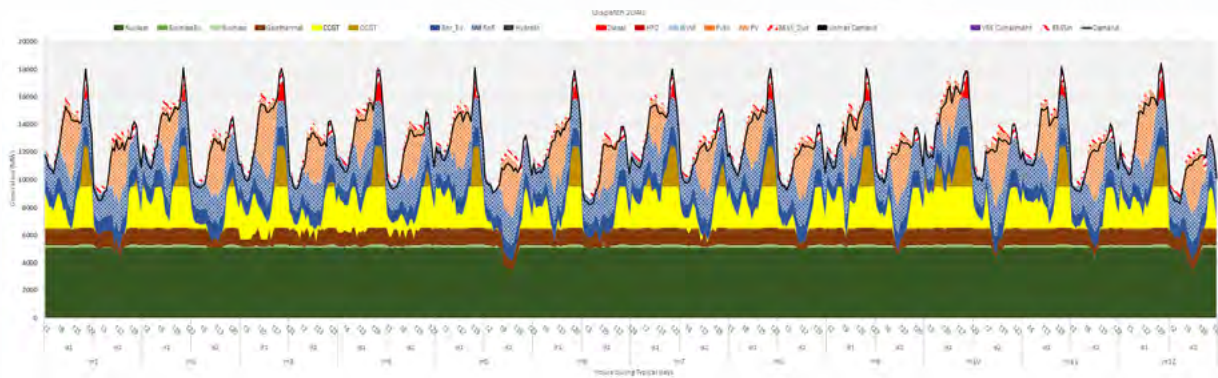


Figure 5-3: 2040 Hourly Dispatch over the representative days each month for the tenfold strategy scenario

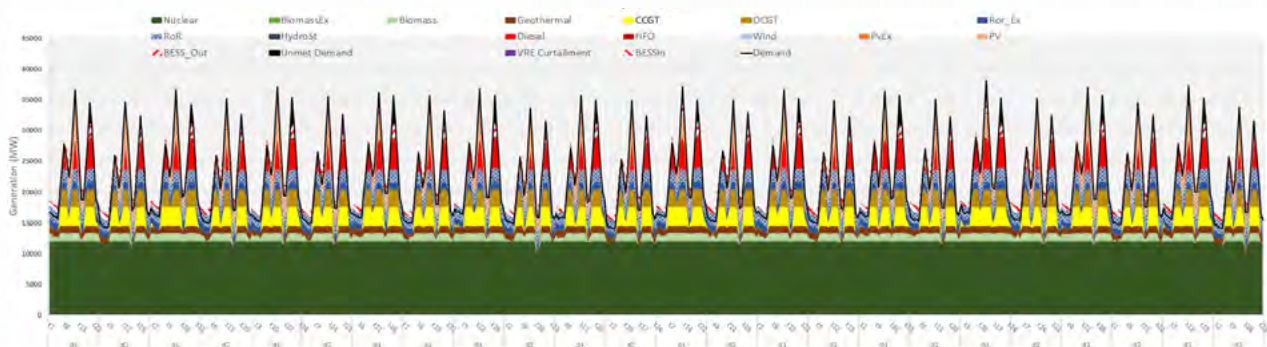


Figure 5-4: 2040 Hourly Dispatch over the representative days each month for the Energy policy 2023 scenario

The model has highlighted that this is the hourly dispatch profiles for the different scenarios in 2040 show that hydropower and gas plants provide crucial flexibility to meet peak demand, especially in BAU and Tenfold scenarios. BESS supports solar

integration by reducing curtailment and shifting energy to peak hours. Nuclear offers steady base load in high-growth scenarios, while solar and wind, though variable, are balanced by storage and fast-ramping technologies. Overall, system reliability depends on a diverse mix of flexible and firm generation.

5.4 COMPARATIVE BENEFITS AND COSTS OF EACH SCENARIO

This section provides a comparative evaluation of the three modeled scenarios Business as Usual (BAU), 10-Fold Strategy, and the Energy Policy 2023 case using key performance indicators including generation output, demand served, capacity addition, investment needs, and renewable energy curtailment. The summary of findings is presented in Table 5-5.

Table 5-5: Summary of the findings over the 2023 - 2040 study horizon.

Parameters	Business As Usual	10-Fold strategy	Energy Policy
VRE curtailment: \$m	15.26	11.44	11.13
NPV of system cost: \$m	2,555.57	14,623.97	30,232.27
Total Generation: GWh	210,862.32	803,952.52	1,359,366.11
Total Demand: GWh	202,513.99	772,562.62	1,269,922.27
Total Capacity Added: MW	2,562.30	25,084.00	39,306.70
Total Investment: \$m	11,144.38	130,421.06	269,236.38
\$/MW	4.35	5.20	6.85

The Total Capacity Added (MW) row refers to the installed generation capacity added over the horizon, and the corresponding Total Investment (\$m) reflects the undiscounted, overnight capital investment required to build the additional generation and transmission capacity specified under each scenario.

Scenario Comparison and Strategic Trade-Offs

- The Business as Usual scenario is the least costly and most conservative, with a system cost NPV of \$2.56 billion and investment needs of \$11.14 billion. However, it results in the lowest generation output and capacity expansion, and carries the risk of constraining Uganda's industrialization, energy access, and clean energy goals.
- The 10-Fold Strategy offers a middle-ground approach with a system cost of \$14.62 billion and investment requirement of \$130.4 billion. It achieves significantly higher generation and demand fulfillment while maintaining moderate curtailment and capacity growth. This scenario requires steady financing and coordinated implementation to realize its full benefits.
- The Energy Policy 2023 scenario is the most ambitious, targeting over 1.36 million GWh of generation, adding over 39,000 MW of generation capacity, and requiring \$269.2 billion in investment. While it has the highest system cost and capital outlay, it supports full energy access, industrial transformation,

and a shift to a clean energy economy. Curtailment is minimized despite the high renewable energy penetration.

Each scenario reflects a strategic trade-off between cost, development impact, and implementation complexity. The Energy Policy scenario, though capital-intensive, provides the most comprehensive path toward achieving Uganda's long-term socio-economic and environmental objectives.

The comparison in Table 5-6 below highlights the strategic trade-offs between the three planning scenarios.

Table 5-6: Scenario Risk-Benefit Comparison

Scenario	Key Risks	Implications (if not implemented)	Key Benefits
Business as Usual	Stalled electrification and industrialization; limits socio-economic progress.	Missed opportunity for economic growth, improved access, and modern energy services.	Low-cost pathway with minimal emissions; maintains status quo with limited investment.
10-Fold Strategy	Requires consistent investment and coordination; risk of underperformance if delayed.	Reduced ability to support large-scale industrial growth and energy access expansion.	Enables scalable growth, better demand satisfaction, and enhanced energy reliability.
Energy Policy 2023	High infrastructure and financing demands; risk of system strain without timely execution.	Failure to achieve universal access, clean energy transition, and regional competitiveness.	Delivers comprehensive sector transformation; aligns with policy, access, and climate goals.

Failure to pursue the higher-growth scenarios could result in missed national targets and lost development opportunities.

5.5 CARBON EMISSIONS AND DECARBONISATION POLICIES

Uganda's Energy Transition Plan (ETP) estimates total national emissions at approximately 115 million metric tons of CO₂ equivalent, with the energy sector accounting for 15% of this total. Notably, electricity generation contributes only about 2%, with a baseline value of 0.345 Mt CO₂eq. As Uganda expands its power generation capacity, tracking the carbon implications of different growth pathways is essential to align with climate goals.

Table 5-7 below presents projected electricity-related emissions under the three planning scenarios across the years 2030 to 2040.

Table 5-7: Projected Electricity-Related Emissions (MtCO₂eq/year)

Scenario	Emissions in MtCO ₂ eq per year		
	2030	2035	2040
BAU	0.24	0.45	0.38
Ten-Fold	0.32	4.98	3.40
Energy Policy	1.27	2.46	14.96

The BAU scenario maintains the lowest emissions across the period due to limited system expansion and minimal fossil fuel reliance. The Ten-Fold Strategy, despite a surge in generation and capacity, remains relatively efficient in emissions by 2040, primarily due to integration of renewables. Conversely, the Energy Policy scenario, while enabling rapid industrialization and clean cooking electrification, leads to significantly higher emissions reaching nearly 15 MtCO₂eq by 2040 highlighting the tradeoff between development scale and carbon intensity.

To further assess Uganda's compliance with the ETP 2030 electricity emissions target of 0.360 MtCO₂eq, Table 5-8 evaluates each scenario's emissions and the associated cost of exceeding (or staying below) the target based on projected carbon pricing.

Table 5-8: Evaluation of Emissions against the ETP targets – 2030

Scenario	Electricity Baseline	ETP2030_Elec_Target	EPM_2030_Emissions_mmt CO ₂ -eq	Violation (mmt CO ₂ -eq)
BAU	0.345	0.360	0.320	-0.040
Tenfold	0.345	0.360	0.179	-0.181
Energy Policy	0.345	0.360	4.180	3.820

The BAU and Ten-Fold scenarios both remain within the 2030 emissions target, with the Ten-Fold strategy achieving the largest margin of compliance and the potential to generate significant savings or carbon credit revenues. In contrast, the Energy Policy scenario exceeds the target by 3.82 MtCO₂eq, resulting in a projected carbon-related cost of over \$1.1 billion, assuming current international carbon pricing benchmarks.

These findings emphasize the importance of climate-aligned energy planning, highlighting the need for low-emission technologies, carbon, and mitigation measures particularly if Uganda pursues the high-growth, high-access pathway under the Energy Policy scenario.

5.6 OBSERVATIONS

The resilience of Uganda's generation and transmission expansion plan is evaluated across multiple dimensions, including technological diversity, system reliability, operational flexibility, and environmental sustainability. Key insights include

- i. Hydropower remains the least-cost generation option, particularly in the short to medium term. However, under the Tenfold Growth and Energy Policy 2023 scenarios, nuclear energy emerges as a strategic base load technology necessary to meet large-scale demand growth and reduce long-term system costs.
- ii. As the energy mix expands to include higher shares of variable renewable energy (VRE) such as solar and wind, the system will require enhanced performance standards. These include spinning reserve, ramp rate capabilities, and grid stability mechanisms, such as Battery Energy Storage Systems (BESS) for PV integration.
- iii. The current transmission network lacks adequate redundancy, particularly in meeting N-1 contingency criteria. While not a direct output of this model, previous analyses and system performance indicate gaps in transmission security and grid resilience under high-demand growth scenarios.
- iv. Demand-side strategies including embedded generation, distributed energy resources, mini-grids, and load management offer opportunities to reshape the load curve, flatten peak demand, and defer large-scale investments, especially under the Energy Policy case.
- v. In the BAU+ and Tenfold scenarios, the model projects a gradual net reduction in emissions, reflecting shifts toward cleaner technologies. However, under the Energy Policy scenario, emissions increase due to the scale of energy use, underscoring the need for affirmative environmental interventions, such as cleaner fuels, carbon pricing, and energy efficiency programs.
- vi. Nuclear development should be fast-tracked, while assessing alternatives such as large hydro, storage, accelerated renewables, and demand-side solutions should be carefully assessed to balance feasibility, financing, and flexibility.
- vii. Regional power trade through the Eastern Africa Power Pool (EAPP) and bilateral interconnections offers an opportunity to diversify supply, optimize the use of regional resources and provide a buffer against demand uncertainty while creating export opportunities for Uganda's surplus generation.

Together, these elements highlight the need for an adaptable, policy-driven implementation approach to ensure the power system remains reliable, secure, and sustainable across all growth trajectories.

6 FINANCING FRAMEWORK AND STRATEGY

Financing is a critical component of any Investment Plan, as it ensures the availability of capital required for the development, construction, operation, and maintenance of the infrastructure. This chapter outlines the key considerations, strategies, and mechanisms for financing the electricity plan, ensuring its feasibility, sustainability, and alignment with broader economic and policy goals.

6.1 SUMMARY COSTS FOR DIFFERENT SCENARIOS

The investment requirements for Uganda's electricity sector vary significantly across the three planning scenarios as seen in Table 6-1; each reflecting a different level of ambition, infrastructure scale, and system complexity.

Table 6-1: Summary of Investment and System Costs by Scenario

Year	SCENARIO		
Costs (USD)	BAU	10 FOLD	ENERGY POLICY
Generation Costs: \$m	7,243.85	115,031.37	246,189.75
Transmission additions: \$m	3,900.53	15,389.68	23,046.63
Total CAPEX Cost : \$m	11,144.38	130,421.06	269,236.38

Under the BAU scenario, an estimated USD 11.14 billion is required to support generation and transmission expansion through 2040. This scenario emphasizes optimization of existing infrastructure and gradual system upgrades. The total annual system cost under this pathway is projected at USD 5.57 billion.

In the 10-Fold Strategy, which supports Uganda's drive to increase economic output tenfold, the investment scale grows substantially. Approximately USD 130.42 billion in capital expenditure will be needed to develop the required generation and transmission assets, with a total annual system cost of USD 35.26 billion. This includes the deployment of over 3,364.9 MW of capacity within fifteen years.

The Energy Policy 2023 scenario presents the most ambitious pathway, aligned with Uganda's targets for universal access, energy transition, and industrialization. It demands USD 269.24 billion in capital expenditure, with an annual system cost reaching USD 71.16 billion. This reflects the substantial infrastructure and generation investments necessary to meet projected demand and policy-driven development goals.

This comparative analysis underscores the scale of investment required to transform Uganda's electricity sector, highlighting the trade-offs between cost, development pace, and national socio-economic objectives.





6.2 LEVELISED COST OF ELECTRICITY

The graph illustrates the Levelised Cost of Electricity (LCOE) for various generation technologies in USD/MWh, breaking down components such as capital costs, O&M, fuel, carbon, and decommissioning annotated with global averages.

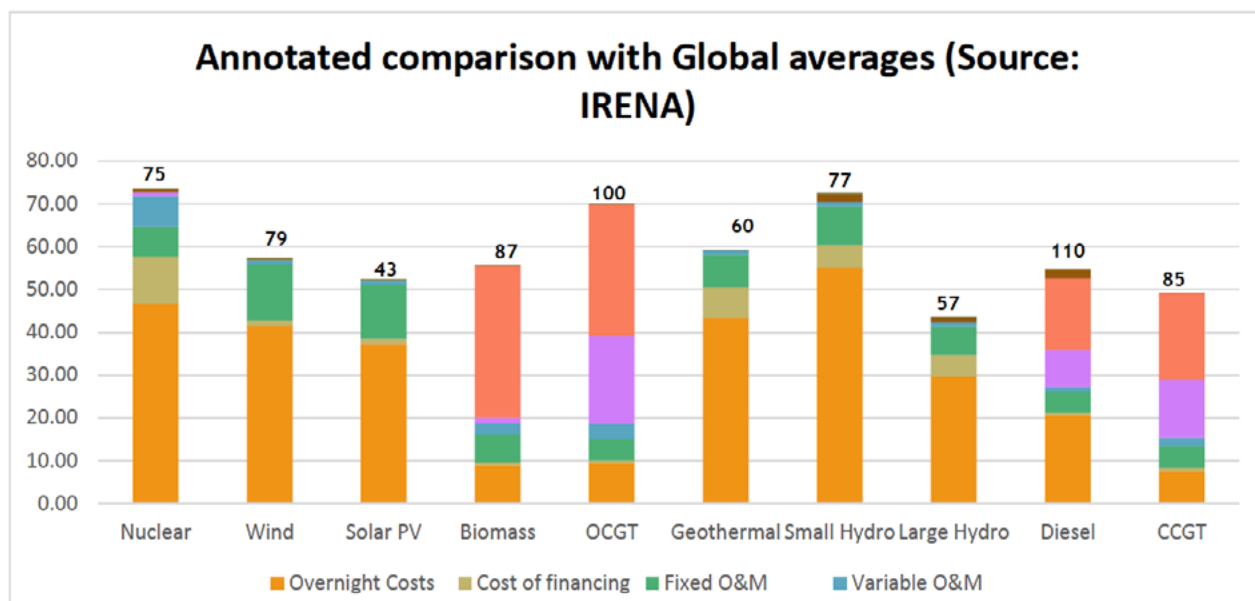


Figure 6-1: Levelized Cost of Electricity (USD/MWh)

Nuclear and small hydro have the highest LCOEs, each around 7.0–7.2 US¢/kWh, mainly due to their high capital and financing costs. In contrast, Large Hydros and CCGT are among the most affordable options, with LCOEs of approximately 4.1 US¢/kWh and 4.9 US¢/kWh, respectively. These values demonstrate the importance of prioritizing lower-cost, flexible technologies while ensuring long-term reliability and sustainability.

The Levelized Cost of Electricity (LCOE), expressed in US\$/kWh, provides a simple and standardized metric for comparing the costs of different generation technologies. While it is widely used for high-level comparisons, LCOE has several important limitations. It does not account for the variability and intermittency of renewable energy sources (RES), relies on assumed rather than actual load factors, and does not capture a power plant's ability to provide capacity at specific times of the day. As such, LCOE alone cannot be relied upon to develop a comprehensive least-cost power system plan.

This limitation becomes even more pronounced in power systems with a high share of Variable Renewable Energy (VRE), such as solar and wind. In such contexts, LCOE fails to reflect the additional integration costs associated with VRE deployment. These include grid expansion requirements, balancing costs due to uncertainty and variability, and profile costs arising from low load factors and limited firm capacity.

Therefore, while LCOE remains a useful benchmark, it should not be used in isolation. A more accurate and holistic assessment of generation technologies particularly in VRE-rich systems requires the inclusion of integration costs to fully capture the system-wide implications for cost, flexibility, and reliability.

6.3 SUMMARY OF FUNDING BY SOURCE

The implementation of Uganda's Least Cost Electricity Generation and Transmission Plan (LCGTP) will require a coordinated financing strategy drawing from both public and private sources. Given the scale of investment required especially under the Energy Policy scenario mobilizing capital from diverse channels will be essential to ensure long-term financial sustainability and project viability. Funding Sources Overview detailed below;

1. Government of Uganda (GoU) Allocations

GoU will continue financing strategic assets, particularly large public infrastructure such as hydropower plants and high-voltage transmission lines. These investments are typically supported through budget allocations for land acquisition, counterpart funding, and equity contributions.

2. Development Partners and Concessional Finance

Concessional loans and grants from bilateral and multilateral partners (e.g., World Bank, AfDB, EU, KfW, JICA) are expected to support both generation and transmission projects. These facilities are generally extended on soft terms, including:

- Grace period: 5–10 years
- Repayment period: 20–30 years and
- interest rate: 0.75–2.0%, often with grant components

3. Private Sector (IPPs and IPTs)

Private capital is expected to finance a growing share of renewable generation and transmission infrastructure, including Independent Power Transmission (IPT) projects. Assumptions for private sector projects include a 70:30 debt-to-equity ratio, with:

- Commercial debt tenor of 10–15 years
- Interest rates ranging 8–12%, depending on risk profile
- Equity Internal Rate of Return (IRR) target of 15–18%.

4. Blended Finance Instruments

To reduce the cost of capital and attract private investment, blended finance will play a key role. Instruments include:

- Viability gap funding (VGF)
- Guarantees and partial risk coverage
- Results-based financing (RBF). These mechanisms are particularly important for VRE and mini-grid projects where commercial viability is challenging.

5. Utility Revenues and Reinvestment

Revenue from electricity sales and internal generation by UEGCL, UETCL, and IPPs will support reinvestment, O&M, and modest network upgrades. However, due to affordability constraints (e.g. industrial tariff caps), full cost recovery is not expected for capital-intensive projects.

6. Carbon and Climate Finance

Funds from climate finance mechanisms such as the Green Climate Fund (GCF), GEAPP, and carbon markets will support clean energy development, particularly for grid flexibility (e.g. BESS), off-grid solar, and clean cooking integration.

This financing structure assumes careful alignment with national debt sustainability thresholds, as guided by the Ministry of Finance and the IMF Debt Sustainability Framework.

6.4 RESOURCE MOBILIZATION STRATEGY

The successful implementation of the Least-Cost Electricity Generation and Transmission Plan (LCGTP) requires a clear strategy to mobilize the financial resources necessary to cover both capital and operational needs, while ensuring resilience to implementation risks. This section outlines the financing requirements, proposed financing strategy, and mechanisms to ensure financial sustainability throughout the planning horizon.

6.4.1 Financing Requirements

The total financing needs under the LCGTP are categorized into three major components: Capital Expenditure (CapEx), Operational Expenditure (OpEx), and Contingency and Risk Mitigation.

6.4.1.1 Capital Expenditure (CapEx)

A. Infrastructure Development

Significant capital investment is required to expand and modernize Uganda's electricity infrastructure in line with the LCGTP. This includes the construction of new generation facilities, expansion of the transmission network, and enhancement of grid reliability through modernization.

- **Generation Plants:** Large-scale investments are needed to develop power generation projects, including the proposed 8,400 MW Buyende Nuclear Power Plant, as well as additional hydro, solar, and geothermal plants to meet long-term energy demand.
- **Transmission Networks:** Expanding and upgrading the national transmission system is essential for evacuating electricity from generation hubs to demand centers particularly in underserved regions such as West Nile and other rural areas where access remains limited.
- **Grid Modernization:** To accommodate the increasing share of Variable Renewable Energy (VRE), investments in Battery Energy Storage Systems (BESS), flexible grid equipment, and advanced control systems are necessary to ensure system stability and operational flexibility.

Funding Sources: Public-private partnerships (PPPs), multilateral development banks (e.g., World Bank, AfDB), green bonds, and government allocations will support these investments.

B. Technology and Equipment

This involves procuring advanced systems for power generation and grid support.

- **Renewable Energy Technologies:** Significant upfront investment

is required for solar panels, wind turbines, and geothermal drilling equipment to expand Uganda's clean energy capacity.

- **Energy Storage Systems:** Deployment of BESS and pumped hydro storage is critical to balance supply and demand, particularly with growing reliance on intermittent renewables.

Funding Sources: This component will be financed through grants, concessional loans, and private sector investments.

C. Land Acquisition and Permitting

This covers costs related to securing land and obtaining regulatory approvals for energy infrastructure.

- **Land Acquisition:** Includes compensation and resettlement for affected communities, especially for large-scale projects like hydropower dams and the Buyende Nuclear Power Plant.
- **Permitting and ESIA:** Encompasses environmental and social impact assessments, stakeholder consultations, and compliance with regulatory standards.

Funding Sources: Primarily supported through government budgets, development partner grants, and project-specific financing.

6.4.1.2 Operational Expenditure (OpEx)

A. Maintenance and Upgrades

Ongoing costs are required to maintain and upgrade power infrastructure to ensure reliability, efficiency, and system longevity.

- **Routine Maintenance:** Covers regular inspections, repairs, and replacements across generation, transmission, and distribution systems.
- **System Upgrades:** Involves modernization efforts such as replacing aging transformers, enhancing substations, and automating grid operations.

Funding Sources: Financed through internally generated funds (IGFs), user tariffs, and government subsidies.

B. Fuel and Resource Costs

These costs cover the procurement and management of inputs required for power generation, especially for thermal and biomass plants.

- **Fuel Procurement:** Includes purchasing and transporting diesel, heavy fuel oil, and natural gas for thermal and gas-fired power plants.
- **Resource Management:** Encompasses the use and monitoring of water for hydropower and biomass for cogeneration.

Funding Sources: These expenses are covered through fuel levies, government subsidies, and operational revenues.

C. Labor and Administrative Costs

This includes all personnel-related and operational management expenses within

the electricity supply industry.

- **Salaries and Benefits:** Covers compensation for staff across generation, transmission, and distribution entities.
- **Training and Capacity Building:** Supports skill development, especially in renewable energy and smart grid technologies.
- **Administrative Overheads:** Includes costs for office operations, stakeholder engagement, and regulatory compliance.

Funding Sources: Financed through operational revenues, government allocations, and donor-supported capacity-building programs.

6.4.1.3 Contingency and Risk Mitigation

A. Reserve Funds

Reserve funds are earmarked to manage unexpected events such as natural disasters, cost overruns, and supply chain disruptions.

- **Project Contingency Funds:** Allocated during planning to cover unforeseen construction or operational costs.
- **Operational Reserve Funds:** Maintained to address revenue shortfalls or emergency maintenance needs.

Funding Sources: Government budgets, development partner grants, and project-specific financing.

B. Insurance

Insurance instruments are critical for protecting energy sector assets and operations from risks.

- **Asset Insurance:** Covers infrastructure like power plants and transmission assets.
- **Liability Insurance:** Protects against third-party claims due to environmental or operational incidents.
- **Business Interruption Insurance:** Compensates for revenue losses during system outages or disruptions.

Funding Sources: Operational budgets, government allocations, and private insurance providers.

6.4.2 FINANCING STRATEGY FOR THE LEAST-COST ELECTRICITY GENERATION AND TRANSMISSION PLAN (LCGTP)

The financing strategy for Uganda's Least-Cost Electricity Generation and Transmission Plan (LCGTP) is designed to mobilize resources from diverse sources to ensure the successful implementation of energy projects.

6.4.2.1 Equity Financing

Equity financing involves raising capital by selling ownership stakes in energy projects. It is a vital mechanism for funding large-scale infrastructure and encouraging private sector participation, especially in capital-intensive ventures.

1. **Private Equity:** In the context of the LCGTP, private investors are expected to play a major role in financing renewable energy projects, bringing not only capital but also technical capacity and innovation, particularly in solar, wind, and geothermal development.
2. **Public Equity:** The government, through SOEs, will contribute equity to strategic national projects such as nuclear and large hydropower plants, ensuring these assets align with national development and energy security goals.
3. **Initial Public Offerings (IPOs):** IPOs will serve as a supplementary mechanism for raising public capital, particularly for established utilities and large-scale energy developers, while promoting transparency and wider ownership in the sector.

6.4.2.2 Debt Financing

Debt financing involves borrowing capital that must be repaid over time, usually with interest. It is well-suited for infrastructure projects with predictable revenue streams, making it a key pillar in financing Uganda's Least-Cost Electricity Generation and Transmission Plan (LCGTP).

1. **Commercial Loans:** These offer immediate capital for infrastructure development, particularly for projects with stable cash flows. In the LCGTP, they are useful for short- to medium-term financing of generation and transmission assets, often through project finance models.
2. **Bonds:** Bonds provide a long-term debt solution for large-scale energy projects, including renewable plants and grid upgrades. The use of green bonds is especially relevant for attracting climate-aligned investment under the LCGTP framework.
3. **Multilateral Development Banks (MDBs):** MDBs such as the World Bank and AfDB offer concessional loans, technical support, and policy guidance. Under the LCGTP, MDBs are critical for de-risking investments in underserved regions and ensuring alignment with Uganda's energy access and transition goals.

6.4.2.3 Public-Private Partnerships (PPPs)

PPPs are cooperative arrangements between public institutions and private sector players to jointly develop, finance, and manage energy infrastructure. They allow for risk-sharing, resource mobilization, and innovation, making them instrumental in delivering the LCGTP's complex and capital-intensive projects.

1. **Collaborative Models:** These include joint ventures where private sector capital and expertise are combined with public oversight to ensure projects align with national priorities. Under the LCGTP, such models are effective for strategic renewable energy and transmission projects requiring high technical input.
1. **Build-Operate-Transfer (BOT):** In BOT arrangements, private entities finance, construct, and operate infrastructure assets for a defined term before handing them over to the government. This model is especially useful for large-scale generation projects, helping reduce upfront public expenditure while ensuring efficiency and timely delivery.

6.4.2.4 Grants and Subsidies

Grants and subsidies offer non-repayable financial support or incentives to reduce the cost burden of energy investments and promote equitable energy access. These instruments are essential for supporting the implementation of the LCGTP, particularly in rural electrification, renewables, and pro-poor energy interventions.

1. **Government Grants:** Provided by national or local governments, these grants fund priority infrastructure, such as rural electrification and renewable energy projects. They help bridge funding gaps and make electricity more affordable for end-users.
2. **International Aid:** Includes grant funding from development partners, international agencies, and donor countries. This support goes toward capacity-building, technology transfer, and infrastructure development, and aligns with climate action and sustainable development goals.
3. **Subsidies:** Offered in the form of feed-in tariffs, tax incentives, or price supports, subsidies encourage investment in clean energy and efficiency by improving financial viability. They also help lower consumer tariffs, particularly in low-income and underserved regions.

6.4.2.5 Crowd funding and Community Financing

Crowd funding and community financing offer innovative, decentralized approaches to mobilize capital for small-scale energy projects. These models are especially relevant for the LCGTP in expanding energy access to rural and off-grid areas.

1. **Community-Based Models:** Involve local communities pooling resources to invest in mini-grids and off-grid renewable systems. These models promote local ownership, ensure alignment with community needs, and foster sustainability through direct stakeholder involvement.
2. **Crowd funding Platforms:** Use online platforms to raise small contributions from a broad base of individuals. This approach supports small-scale renewable energy projects, while increasing public engagement and awareness of sustainable energy development.

6.4.3. Financial Instruments and Mechanisms

To mobilize funding and ensure sustainable implementation of the LCGTP, a range of financial instruments and mechanisms can be employed. These tools enhance investor confidence, promote climate-friendly investments, and ensure financial sustainability across the energy value chain.

6.4.3.1 Green Bonds

Green bonds are debt instruments issued specifically to finance projects with positive environmental impacts, such as renewable energy, energy efficiency, and climate resilience.

In the context of the LCGTP, green bonds can fund large-scale renewable energy projects like solar and wind farms, as well as grid modernization efforts including BESS and smart grids. Their structure appeals to environmentally conscious investors and development partners, offering a mechanism that ensures transparency and accountability through strict reporting on fund utilization and project impacts. They are instrumental in attracting long-term, climate-aligned capital to Uganda's energy sector.

6.4.3.2 Carbon Credits and Emissions Trading

Carbon credits and emissions trading are market-based mechanisms that reward the reduction of greenhouse gas emissions. Each carbon credit represents the offset of one ton of CO₂ equivalent, which can be traded on global carbon markets.

Within the LCGTP, these instruments offer an additional revenue stream for renewable and energy efficiency projects. The proceeds can be reinvested into clean energy initiatives, supporting Uganda's transition to a low-carbon economy. Furthermore, carbon markets enhance Uganda's ability to meet international climate commitments and encourage innovation in low-emission technologies and practices.

6.4.3.3 Energy Service Companies (ESCOs)

ESCOs are specialized businesses that deliver energy efficiency solutions, typically under performance-based contracts where they are paid based on the actual energy savings achieved.

In the context of the LCGTP, ESCOs can play a vital role in reducing energy demand and operational costs across buildings, industries, and public infrastructure. Their model encourages private sector investment in energy efficiency while promoting cost savings for end-users, particularly in low-income communities. By linking payments to verified savings, ESCOs also ensure accountability and effective implementation.

6.4.3.4 Revolving Funds

Revolving funds are self-sustaining financing mechanisms where repayments or returns from earlier investments are recycled to finance new projects, creating a continuous capital flow.

In the LCGTP framework, these funds offer sustainable financing, especially for small-scale renewable energy projects like mini-grids and solar home systems in rural areas. They help leverage initial investments, allowing the fund to expand over time, and promote local ownership and accountability by empowering communities to manage and reinvest in their own energy solutions.



Returns from earlier investments are recycled to finance new projects, creating a continuous capital flow.

7 IMPLEMENTATION ARRANGEMENTS

7.1 ARRANGEMENTS FOR COORDINATING MDA PLAN IMPLEMENTATION

The successful execution of the Least Cost Generation and Transmission Plan (LCGTP) requires effective coordination among MDAs across the Electricity Supply Industry (ESI).

Table 7-1 below outlines the institutions involved and their respective roles in delivering and supporting the plan's implementation.

Table 7-1: Implementing agencies of the plan of the Least Cost Generation Transmission Plan.

No.	Institution	Role
	Ministry of Energy and Mineral Development	Planning, Policy guidance and Investment promotion.
	Ministry of Finance, Planning and Economic Development	Investment financing and public sector resource allocation
	Electricity Regulatory Authority	Sector regulation, licensing, tariff approval
	Uganda Electricity Generation Company Ltd (UEGCL)	Construction, maintenance, and operation of public generation plants
	Uganda Electricity Transmission Company Ltd (UETCL)	System Operator and transmission developer. Manages the high voltage electricity transmission infrastructure the country.
	Uganda Electricity Distribution Company Ltd (UEDCL)	Manage the entire electricity distribution network across Uganda at and below 33 kV
	National Planning Authority	Coordination of national development frameworks and energy planning alignment
	Uganda Investment Authority.	Promotion and facilitation of domestic and foreign investment in the power sector

To ensure inter- and intra-sectoral coordination, the implementation of the LCGTP will be overseen through routine joint technical meetings, planning alignment forums, and harmonized progress monitoring across agencies. Sector-wide engagement will be facilitated through the Energy Sector Working Group and aligned to national budgeting and policy cycles.

A dedicated Core EPM Modelling Team shall be institutionalized and hosted within MEMD. The team will include representatives from MEMD, ERA, UEGCL, and UETCL, and will be responsible for the routine review, update, and validation of the Plan using the Energy Planning Model (EPM). Supported under the Energy Sector Planning and Coordination Committee (ESPCC), the team will meet quarterly and produce an annual update of the Plan to incorporate emerging developments within the electricity sector.

7.2 SUSTAINABILITY ARRANGEMENTS

To ensure the long-term sustainability of the LCGTP, the Plan will be supported by the following pillars:

- **Institutional Sustainability:** The EPM planning framework will be embedded within MEMD and supported by a permanent cross-agency modelling team. This ensures continuity of expertise and institutional memory.
- **Programmatic Sustainability:** Routine planning updates and scenario revisions will be integrated into the sector's annual planning cycle, ensuring alignment with national targets, sector investment plans, and utility forecasts.
- **Financial Sustainability:** The planning function will be funded under government budget allocations and supported by development partners through the ESPCC. This will ensure the availability of resources for technical tools, data updates, training, and stakeholder consultation.

The Core EPM Modelling Team will play a central role in maintaining the integrity of the planning process. It will include experts from across the ESI and be equipped to continuously improve the model's assumptions, structure, and outputs to reflect Uganda's evolving energy landscape.



8 COMMUNICATION AND STAKEHOLDER MANAGEMENT STRATEGY

8.1 OVERVIEW

Effective communication and stakeholder engagement are critical for the successful adoption and implementation of the 52GW Generation-Transmission Least-Cost Plan. This strategy outlines the approach to sharing key findings, gathering feedback, and ensuring stakeholder buy-in from relevant institutions, including the Ministry of Energy and Mineral Development, Ministry of Finance, Planning and Economic Development, Electricity Regulatory Authority (ERA), National Planning Authority, Uganda Investment Authority, Uganda Electricity Generation Company Limited (UEGCL), Uganda Electricity Transmission Company Limited (UETCL) and Uganda Electricity Distribution Company Limited (UEDCL).

8.2 OBJECTIVES OF THIS COMMUNICATION AND STAKEHOLDER MANAGEMENT STRATEGY

The objectives of this communication and stakeholder management strategy are;

- i. Create awareness about the least cost Plan amongst the stakeholders.
- ii. Identify and map out key stakeholders and their roles in implementation of the least cost plan.
- iii. To disseminate the findings and recommendations of the 52GW least-cost plan to all stakeholders.
- iv. To ensure transparency and inclusivity in the decision-making process by providing all relevant stakeholders with timely, accurate, and accessible information. Open dialogue and structured engagement sessions will foster trust and collective ownership of the plan.
- v. To facilitate constructive feedback and incorporate stakeholder inputs for refinement of this least cost plan.
- vi. To build consensus and foster collaboration among the stakeholders for effective plan implementation.

8.3 COMMUNICATION AND FEEDBACK METHODS

The following table outlines key stakeholders and proposed communication methods.

Table 8-1: Key Stakeholders, Roles, and Communication Channels in the Project

Stakeholder	Role in the Project	Communication and Feedback Methods
Ministry of Energy and Mineral Development	Policy formulation and approval	High-level meetings, policy briefs, workshops and reports

Ministry of Finance, Planning and Economic Development	Budget allocation and financial oversight	Financial reports, consultations, workshops and budget meetings
UEGCL	Data provision on Generation operations and planning	High level meetings, Technical consultations, reports and workshops
UETCL	Data provision on Transmission planning and operations	High level meetings, Technical consultations, reports and workshops.
UEDCL	Data provision on Distribution network management	High level meetings, Technical consultations, reports and workshops.
Uganda Investment Authority	Investment promotion and funding mobilization	Investment forums and strategic meetings
National Planning Authority	Strategic national development planning	Policy alignment discussions, technical reports and presentations



9 RISK MANAGEMENT

To ensure the successful implementation of the LCGTP, the Ministry of Energy and Mineral Development (MEMD) has applied its risk management framework to evaluate the potential risks.

The plan aims to proactively mitigate these risks by strengthening infrastructure resilience, addressing institutional gaps, and implementing a risk-informed approach. Achieving the plan's objectives ensuring a least-cost generation and transmission mix, security of supply, and increased access to electricity requires continuous identification, mitigation, and monitoring of potential risks.

Below is an outline of the risks associated with the LCGTP, categorized as High (H), Medium (M), and Low (L), along with mitigation measures and strategies to effectively manage them.

9.1 THE RISK MAPPING MATRIX

The Risk Assessment Matrix helps in evaluating the severity and likelihood of each risk and prioritizing them for mitigation based on their likelihood and impact. The Likelihood and Impact are typically rated on a scale from 1 to 5:

Likelihood:

1 = Rare, 2 = Unlikely, 3 = Possible, 4 = Likely, 5 = Almost Certain

Impact:

1 = Minor, 2 = Moderate, 3 = Major, 4 = Critical, 5 = Catastrophic

Table 9-1: LCGTP Risk assessment

Risk	Likelihood (1-5)	Impact (1-5)	Risk Level (Impact x Likelihood)	Risk Category	Mitigation Strategy
Insufficient Funding and Budget Constraints	4	5	20	Institutional & Financial	Advocate for increased funding, diversify revenue sources, strengthen financial management
Aging and Inadequate Energy Infrastructure	4	5	20	Infrastructure	Invest in modernization, prioritize renewable energy, improve maintenance
Technological Failures and Security Risks	3	4	12	Technological	Invest in cyber security, adopt advanced technology, provide regular training

Weak Policy Enforcement	3	3	9	Regulatory & Policy	Strengthen enforcement, regular audits, capacity building for regulatory bodies
Resistance to sector Projects	3	3	9	Public Resistance	Conduct outreach, involve communities in decision-making, ensure fair compensation

The LCGTP faces high-risk challenges primarily in funding constraints, aging infrastructure, and environmental concerns, including heavy nuclear deployment. Addressing these risks is crucial to ensuring the plan’s success.

Insufficient funding and budget constraints threaten the implementation of critical projects. To mitigate this, the Ministry will advocate for increased funding, diversify revenue sources, and strengthen financial management to ensure sustainable investment in the energy sector.

Aging and inadequate energy infrastructure pose a significant risk to reliability and efficiency. The strategy involves modernizing the grid, prioritizing renewable energy investments, and enhancing maintenance practices to improve system resilience.

By addressing these high-risk areas, the MEMD aims to secure sustainable financing, enhance infrastructure resilience, and ensure environmental and regulatory compliance for the successful execution of the LCGTP.

9.2 RISK CATEGORISATION

The Least Cost Generation and Transmission Plan (LCGTP) 2025/26 – 2039/40 operates within a complex risk landscape, categorized into operational, strategic, and external risks.

Based on the ratings above in section 13.1; risks are categorized into different levels:

- High (16-25): Immediate action required, prioritize mitigation
- Medium (9-15): Moderate action, monitor and plan mitigation
- Low (1-8): Monitor but minimal action needed at present

Table 9-2: Risk levels for the different categories.

SN	Risk Category	Risk	Risk factor	Risk Level	Mitigation
1	Operational	Institutional and financial	Limited funding due to budget cuts, weak financial management systems, corruption	High	Strengthen financial controls, enhance accountability mechanisms, Engage MoFPED on budget allocation.
2	External risk	Market and economic instability	Fluctuations prices	High	Diversify revenue sources
		Environmental and climate risks	Climate change effects, natural disasters, environmental degradation	High	Strengthen environmental policies, invest in climate-resilient infrastructure, promote sustainable energy solutions
3	Strategic risk	Policy and regulatory risks	Inconsistent policies, regulatory bottlenecks, lack of enforcement	Medium High	Regular policy reviews, stakeholder engagement, capacity building for regulators
		Public resistance to MEMD projects	Land acquisition conflicts, lack of stakeholder buy-in	Medium	Community sensitization, fair compensation policies, transparent engagement processes

10 MONITORING AND EVALUATION FRAMEWORK

10.1 OVERVIEW

The monitoring and evaluation (M&E) framework for the 52GW Generation-Transmission Least-Cost Plan is designed to track progress, assess performance, and ensure alignment with Uganda's Energy Policy (2023). The M&E framework will provide a structured approach to performance monitoring, decision-making, and policy adjustments to achieve national energy objectives.

This M&E shall cover progress reporting, MDA performance annual review, mid-term evaluation and end of term evaluation.

10.2 MONITORING AND EVALUATION OBJECTIVES

1. To track progress, assess performance, and evaluate the effectiveness of the LCGTP in achieving its objectives.
2. To track progress in implementing the least-cost generation-transmission plan by tracking key performance indicators.
3. To assess the effectiveness and efficiency of energy policy interventions.
4. To ensure informed decision-making through periodic performance evaluations.
5. To enhance accountability and transparency in the energy sector.
6. To provide feedback for continuous improvement of energy programs and strategies.

10.3 MONITORING AND EVALUATION FRAMEWORK

The M&E framework is structured around the results chain which looks at the input, output, outcome, and impact indicators to assess performance and success in the implementation of the least cost generation and transmission plan

A. Key Performance Indicators

The framework aligns with Uganda's Energy Policy tracking framework and includes the indicators hereunder:

1. Access to Electricity: Increase in the number of households and businesses with access to electricity.
2. Renewable Energy Integration: Percentage of renewable energy sources in the energy mix.
3. Energy Efficiency: Reduction in energy losses and improvement in energy efficiency.
4. Financial Sustainability: Improvement in the financial performance of the

energy sector.

5. Operational Sustainability: Improvement in the operational performance of the energy sector.

B. Output Indicators:

1. Number of new electricity connections (Target: 500,000 new connections by 2030).
2. Capacity of new generation plants (Target: 1,000 MW of new generation capacity by 2030).
3. Length of new transmission lines (Target: 1,500 km of new transmission lines by 2030).

C. Outcome Indicators:

1. Increase in electricity access (Target: 80% of households and businesses with access by 2030).
2. Reduction in energy poverty (Target: 50% reduction by 2030).
3. Improvement in energy security (Target: 90% reliability of electricity supply by 2030).

D. Impact Indicators:

1. GDP growth (Target: 7% annual GDP growth by 2030).
2. Poverty reduction (Target: 50% reduction in poverty by 2030).
3. Improved health and education outcomes (Target: 20% improvement by 2030).

10.4 PROGRESS REPORTING MECHANISMS

The M&E framework establishes structured reporting mechanisms at different intervals to provide timely updates on implementation progress:

1. **Quarterly Progress Reports:** Submitted to the Ministry of Energy and Mineral Development (MEMD).
2. **Annual Progress Reports:** Submitted to MEMD and the National Planning Authority (NPA), including financial updates and revised performance indicators.
3. **MDA Performance Annual Review:** Conducted by MEMD to assess institutional performance in implementing the plan.
4. **MDA Performance Report:** Submitted to the Ministry of Energy and Mineral Development and the National Planning Authority.
5. **Sector Performance Reports:** Published to communicate progress by the respective implementation agency i.3e challenges, and recommendations for energy sector stakeholders.

10.5 EVALUATION MECHANISMS

Evaluation is a key component of the M&E strategy and will be conducted at three main levels:

1. Mid-Term Evaluation (2030): Conducted by an independent consultant to assess progress, challenges, and necessary policy adjustments.
2. Mid-Term Evaluation Recommendations: Implemented by the Ministry of Energy and Mineral Development and sector stakeholders.
3. End-of-Term Evaluation (2035): A comprehensive review of policy impact, effectiveness, and sustainability, conducted by an independent consultant.
4. End of Term Evaluation Recommendations: Implemented by the Ministry of Energy and Mineral Development and sector stakeholders.
5. Value-for-Money Audits: Led by the Office of the Auditor General to ensure efficient use of resources in policy implementation.

10.6 DATA COLLECTION AND PERFORMANCE REVIEW

There will be continuous data collection and performance review of the implementation of the plan. Below are some of the key areas of focus.

1. Baseline data will be established at the onset to track progress.
2. National energy statistics will be updated annually and integrated into the energy sector database.
3. The MEMD, in collaboration with stakeholders, will collect and analyse energy sector data for policy review.

10.7 POLICY REVIEW AND CONTINUOUS IMPROVEMENT

The policy review process will ensure adaptability and responsiveness to emerging trends. It includes:

1. Periodic policy reviews every 3-5 years to align with evolving national and international energy dynamics.
2. Stakeholder consultations to incorporate feedback and address emerging challenges.
3. The development of revised implementation work plans based on M&E findings.

11 RECOMMENDATIONS

1. Prioritize development of projects based on LCGTP insights

Project selection should be anchored in the Least-Cost Generation and Transmission Plan to ensure investments deliver maximum value. The LCGTP provides evidence-based sequencing of generation and transmission projects, helping avoid stranded assets, manage costs, and align system expansion with national energy and industrialization goals.

2. Undertake comprehensive feasibility studies for all large power projects

All major power projects must undergo full technical, economic, environmental, social, and financial assessments before implementation. Comprehensive feasibility studies improve bankability, reduce delays, and ensure that projects are well-prepared, cost-effective, and compatible with system needs.

3. Operationalize and capitalize the Electricity Development Fund (EDF)

There is an urgent need to activate and adequately fund the EDF to support strategic sector investments. A functional EDF will strengthen project preparation, unlock financing for critical infrastructure, and help manage affordability through targeted support for grid reinforcement and access initiatives.

4. Fast-track development of additional base-load sources

Timely development of new base-load capacity is essential to sustain reliability and support industrialization. Accelerating these projects will help meet growing demand, stabilize the grid as renewable penetration increases, and reduce reliance on costly emergency or diesel-based generation.

5. Screen large load customers to manage grid and tariff risks

Large industrial loads should be systematically evaluated to understand their impact on grid performance and system costs. Proper screening enables tailored connection solutions, protects the network from overloads, and ensures that new customers contribute fairly to network reinforcement needs.

6. Regulate adoption of intermittent generation technologies

Stricter controls are needed to guide the integration of solar, wind, and other variable renewables. Clear technical standards, grid-impact studies, and firm operational requirements will safeguard grid stability while enabling orderly and sustainable renewable energy expansion.

7. Develop a national grid flexibility and resilience strategy

A comprehensive flexibility and resilience plan should guide investments that strengthen Uganda's grid for future challenges. This strategy will define the role of storage, flexible generation, interconnections, and system operations enhancements in supporting reliability, renewable integration, and climate resilience.

12 REFERENCES

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12 APPENDIX

12.1 EXISTING GENERATION PLANTS

Project Name	Region	Capacity	Starting year
Bujagali	JINJA_MET-RO	250	2012
Karuma	NORTH	600	2021
Isimba	JINJA_MET-RO	183	2019
Kiira HPP	JINJA_MET-RO	200	2000
Nalubale HPP	JINJA_MET-RO	180	1954
Mobuku 1 (KML)	WEST	5	2005
Mobuku 3 (KCCL)	WEST	9.9	2003
Ishasha	WEST	6.4	2011
Bugoye	WEST	13	2009
Mpanga	WEST	18	2011
Buseruka Hydromax	WEST	9	
Kikagati	WEST	14	2021
Rwimi	WEST	5.54	2017
Waki	WEST	4.8	2018
Lubilia	WEST	5.4	2018
Siti 1	EAST	5	2017
Siti 2	EAST	16.5	2019
Nyamwamba	WEST	9.2	2018
Nyamwamba II	WEST	7.8	2022
Achwa I	NORTH	46.2	2021
Achwa II	NORTH	41	2019
Sindila	WEST	5.25	2019
Ndugutu	WEST	5.9	2019
Muyembe-Sirimityo	EAST	6.9	2021
Kakaka/Greenewus energy	WEST	4.5	2021
Kyambura	WEST	7.6	2019
Muvumbe	WEST	6.5	2017
Mahoma	WEST	2.7	2018
Nkusi HPP	WEST	9.6	2018
Nyamagasani 2 HPP	WEST	6.1	2018
Nyamagasani 1 HPP	WEST	15	2021

Bukinda HPP	WEST	6.5	2020
Nyagak	WEST_NILE	3.5	2014
Kisizi_offgrid	WEST	0.38	2017
Bwindi_offgrid	WEST	0.06	2018
Kuluva hospital_offgrid	WEST_NILE	0.12	2023
Nyagak III	West Nile	5.6	2025
Kigwabya	West	4.2	2024
TOTAL ROR		1710.35	
Namanve	CENTRAL	50	2009
Electromaxx	EAST	50	2009
Electromax- Arua off-grid	WEST_NILE	8	
TOTAL THERMAL POWER		108	
Tororo Solar North	EAST	10	2017
Access Solar TSK	EAST	10	2017
Xsabo solar Kabulasoke	CENTRAL	20	2018
Emerging Power Bufulu-bi	EAST	10	2019
Nkonge Xsabo solar	CENTRAL	20	2023
Tororo PV Power Co Ltd	EAST	10	2019
Busitema	EAST	4	2022
Sagewood_Offgrid	WEST	0.17	2021
Bunjako_Offgrid	CENTRAL	0.12	2022
Absolute - Kitobo_Offgrid	CENTRAL	0.23	2018
KIS_Offgrid	CENTRAL	1.6	2017
Lamwo_Offgrid	NORTH	1	2022
Bukasa Island _Offgrid	CENTRAL	0.1	2020
Nsambya_offgrid	CENTRAL	0.06	2022
Katiirwe_Offgrid	WEST	0.06	2022
Bussi_Offgrid	CENTRAL	0.4	2023
Lolwe_Offgrid	CENTRAL	0.6	2022
TOTAL PV		88.34	
SCOUL	CENTRAL	25	2018
Sugar Allied Ind	EAST	11.9	2015
Kakira Sugar	JINJA_MET-RO	51.1	2009
Mayuge	JINJA_MET-RO	23	2015
GM Sugar	JINJA_MET-RO	7.35	2021

Kamuli Sugar	JINJA_MET-RO	3	2021
GOTNUR_Offgrid	NORTH	0.44	2022
Kinyara Sugar	WEST	14.5	2010
Hoima Sugar	WEST	12	2023
Pamoja_Tiribogo_Off-grid	CENTRAL	0.03	2018
Pamoja_Ssekanyonyi_Offgrid	CENTRAL	0.01	2018
TOTAL BIOMASS		148.33	

13.2 COMMITTED GENERATION PROJECTS

Project Name	Region	Capacity (MW)	Commercial Operation Date (COD)
Hoimo	West	3.3	2027
Igassa	West	0.3	2027
Nsongya	West	0.7	2027
Nyabuhuka	West	3.2	2026
Nyamabuye	West	7	2028
Tokwe	West	0.3	2027
Kabasanja	Central	0.4	2027
Katooke	Central	0.3	2027
Kabeywa 1	East	6.5	2028
Kabeywa 2	East	2	2029
Nchwera	West	0.5	2027
Simu	East	9.5	2027
Sisi	East	7	2027
Warugo	West	0.5	2028
TOTAL ROR		51.3	
Rupa wind		20	2025
TOTAL WIND		20	
SCOUL Bagasse	Jinja Metro	9.5	2025
TOTAL BIOMASS		9.5	
Ituka Solar PV plant	West Nile	20	2026
TOTAL PV		20	

13.3 CANDIDATE GENERATION PROJECTS

Project Name	Region	Capacity	Starting Year
Ayago	West Nile	840	2032
Oriang	North	392	2033
Kiba	West Nile	400	2031
Uhuru	North	390	2036
Achwa Aber	North	52	2029
Agago-Achwa IV	North	10.5	2029
Atari	East	3.3	2028
Awere	North	18	2029
Buwangani	East	7	2028
Kiraboha	West	5	2027
Kisinga	West	2.5	2028
Latoro	West Nile	4.2	2030
Lower Achwa	North	17.4	2029
Lwakhakha	East	6.7	2030
Maziba	West	1.2	2027
Mini hydros 3	East	19	2031
Mitano	West	13.6	2027
Mukoki	West	3.4	2028
Muzizi	West	50	2030
Nengo Bridge	West	7.5	2027
Ngenge	West	13.8	2030
Nshongezi	West	39.0	2027
Nsongi	West	7	2027
Nyakinengo	West	5.2	2028
Okollo	West Nile	5	2029
Rwembya	West	0.4	2030
Sironko	East	7.0	2027
30 mini hydros	West	180	2029
Total ROR		2500.7	
Central PV	Central	600	2030
East PV	East	5800	2030
North PV	North	1150	2030
West Nile PV	West Nile	1610	2030
West PV	West	437	2030
Jinja Floating	Jinja Metro	270	2030
Total PV		9,867	
Senok Wind		20	2026
Karamoja Wind	East	120	

Wind Generic	West Nile	80	2026
Total Wind		220	2036
Nakasongola	Central	7200	2036
Buyende	East	8400	2032
Lamor	North	4000	2046
Kiruhura	West	4800	2041
Total Nuclear		24,400	
Biogas Power Plant Masindi	West	20	2028
Pece Biomass	North	20	2028
Victoria Sugar		10	
Kyankwanzi Ba-gasse	West	9	2028
Unergy Bio	Central	20	2028
Central Bio	Central	200	2026
Jinja Wte	East	2.5	2026
Kamuli Sugar Ba-gasse	East	3	2027
Pramukh Bagasse	East	8	2026
Pece	North	20	2027
Kabale Peat	West	30	2029
Kinyara Sugar	East	30	2027
Kyoga Peat	West	50	2028
West_Biomass	West	620	2028
East Biomass	East	461	2028
Total Biomass		1,483.5	
Total Bess		1,551.32	
Panyimur		150	2027
Katwe		680	2030
Kibiro		250	2028
Elegu		90	
Buranga		380	2028
Total Geothermal		1,550	
East CCGT	East	1250	2030
West CCGT	West	1500	2030
Total CCGT		2,750	
East OCGT	East	1250	2030
West OCGT	West	1646	2030
Total OCGT		2,896	

13.4 INSTALLED CAPACITY BY TECHNOLOGY FOR DIFFERENT SCENARIOS

13.4.1 BAU SCENARIO

No	Technologies	2023	2030	2035	2040	%Share
	Wind		20.00	20.00	20.00	0.5%
	PV	85.73	85.73	316.10	561.03	13.5%
	HFO	100.20	100.20	100.20	100.20	2.4%
	Biomass	148.85	158.35	158.35	158.35	3.8%
	CCGT			200.00	200.00	4.8%
	OCGT		39.10	463.28	463.28	11.2%
	RoR	1,730.96	1,788.36	2,075.23	2,638.26	63.7%
	BESS			424.83	461.50	
	TOTAL	2,065.74	2,391.74	3,757.99	4,602.63	

13.4.2 TEN FOLD GROWTH SCENARIO

No.	Technologies	2023	2030	2035	2040	%Share
	Nuclear				5,030.01	19.6%
	Wind		20.00	220.00	220.00	0.9%
	PV	85.73	580.31	5,912.15	6,922.15	27.0%
	HFO	100.20	100.20	100.20	100.20	0.4%
	Diesel			191.37	1,691.37	6.6%
	Geothermal		727.76	1,210.00	1,210.00	4.7%
	Biomass	148.85	158.35	158.35	158.35	0.6%
	CCGT		461.02	2,950.00	2,950.00	11.5%
	OCGT		736.98	2,668.63	2,935.10	11.5%
	RoR	1,730.96	3,193.06	4,377.06	4,408.26	17.2%
	BESS			1,500.00	1,500.00	
	TOTAL	2,065.74	5,977.68	19,287.76	27,125.44	

ENERGY POLICY SCENARIO

No.	Technologies	2023	2030	2035	2040	%Share
	Nuclear			4,800.00	12,000.00	31.3%
	Wind		20.00	20.00	220.00	0.6%
	PV	85.73	2,937.58	6,115.73	8,052.73	21.0%
	HFO	100.20	100.20	100.20	100.20	0.3%
	Diesel		1,000.00	3,500.00	5,000.00	13.0%
	Geothermal		1,210.00	1,360.00	1,360.00	3.5%

	Biomass	148.85	158.35	658.35	1,321.85	3.4%
	CCGT		2,042.17	2,042.17	2,950.00	7.7%
	OCGT		170.25	1,435.10	2,935.10	7.7%
	RoR	1,730.96	3,193.06	3,993.06	4,408.26	11.5%
	BESS			3,000.00	3,000.00	
	TOTAL	2,065.74	10,831.61	27,024.61	41,348.14	

13.5 SHARE OF TOTAL ELECTRICITY DEMAND ATTRIBUTABLE TO KEY DEMAND DRIVERS UNDER THE ENERGY POLICY SCENARIO, 2024-2040

Demand drivers	2024	2025	2026	2027	2028	2029	2030	2031	2032	2033	2034	2035	2036	2037	2038	2039	2040
BAU	48.39%	43.16%	45.75%	26.97%	24.13%	26.66%	22.71%	21.18%	19.73%	18.25%	16.77%	15.29%	13.86%	12.50%	11.23%	11.31%	11.30%
Power Exports	5.51%	5.46%	5.46%	3.04%	2.56%	6.60%	13.71%	11.93%	10.36%	8.93%	7.65%	6.49%	5.48%	4.60%	3.85%	3.60%	4.64%
Data centres	0.65%	2.78%	8.35%	18.59%	15.70%	12.68%	10.08%	9.26%	8.04%	6.93%	5.94%	5.04%	4.26%	3.57%	2.99%	2.80%	2.60%
Biomass conversion	38.96%	40.62%	50.34%	34.71%	36.31%	33.61%	33.10%	35.69%	38.41%	41.03%	43.53%	45.85%	47.95%	49.88%	51.67%	48.88%	46.80%
Industrial Parks	0.65%	0.55%	0.55%	0.61%	0.64%	0.57%	0.55%	0.71%	0.93%	1.20%	1.55%	1.97%	2.49%	3.14%	3.94%	5.53%	6.50%
E-mobility	4.87%	6.28%	10.85%	11.63%	15.66%	14.90%	14.63%	15.52%	16.32%	16.96%	17.48%	17.88%	18.09%	18.00%	17.59%	18.19%	18.56%
SGR, Oil pipeline, Mining	0.00%	0.00%	3.28%	2.73%	3.08%	2.76%	2.63%	2.74%	2.86%	2.96%	3.04%	3.10%	3.14%	3.16%	3.17%	3.00%	2.78%

13.6 GENERATOR PLANT DATA SUMMARY: CAPACITY, COSTS, AND OPERATING PARAMETERS

Plants	* Comm year	* In- stalled capac- ity	* Plant Sta- tus	* Primary fuel	Heat rate	Fixed O&M	Variable O&M	Capex	Min gener- ator load- ing	Maximum contribution to reserve	Capacity Credit
		MW			MMBTu/MWh	\$/MW/year	\$/MWh	\$/m/MW	%	%	MW
Plants	StYr	Capac- ity	Status	fuel1	HeatRate	FOMper- MW	VOM	Capex	MinLimit- Share	ResLimShare	Capacity Credit
Xsabo_PV	2018	20.0	Existing	11.PV_Ex		16,351	1.0				0.03
BukasaOffgrid_PV	2020	0.1	Existing	11.PV_Ex		10,000	1.0				0.03
KISs_PV	2015	0.6	Existing	11.PV_Ex		10,000	1.0				0.03
Kitobo_PV	2016	0.2	Existing	11.PV_Ex		10,000	1.0				0.03
Nkongwe_PV	2023	20.0	Existing	11.PV_Ex		10,000	1.0				0.03
Pam_Ssekanyonyi_PV	2014	0.1	Existing	11.PV_Ex		10,000	1.0				0.03
Pam_Tiribogo_PV	2014	0.1	Existing	11.PV_Ex		10,000	1.0				0.03
Jacobsen_TPP	2008	50.2	Existing	13.HFO	8.76	40,227	4.0				1.00
SCOUL	2020	25.0	Existing	3.Biomass_ Ex	16.88	30,000	4.0		40%		1.00
Lolwe_BIO	2022	0.6	Existing	3.Biomass_ Ex	16.88	30,000	4.0		40%		1.00
BUNJAKO	2022	0.1	Existing	7.RoR_Ex		40,000	1.0				1.00
Access_PV	2016	10.0	Existing	11.PV_Ex		12,828	1.0				0.03
Busitema_PV	2022	4.0	Existing	11.PV_Ex		10,000	1.0				0.03
EmergingPowerPV	2019	10.0	Existing	11.PV_Ex		9,771	1.0				0.03
Tororo_PV	2019	10.6	Existing	11.PV_Ex		78,841	1.0				0.03
TororoNorth_PV	2017	10.0	Existing	11.PV_Ex		26,632	1.0				0.03
Electro_Maxx_TPP	2012	50.0	Existing	13.HFO	8.76	25,382	4.0				1.00
Sugar_Allied_Ind_BIO	2015	11.9	Existing	3.Biomass_ Ex	12.50	30,000	4.0		40%		1.00
Siti_I_HPP	2017	6.1	Existing	7.RoR_Ex		123,372	1.0				1.00
Siti_II_HPP	2019	16.5	Existing	7.RoR_Ex		64,848	1.0				1.00
Muyembe_HPP	2021	7.2	Existing	7.RoR_Ex		30,169	1.0				1.00
Kakira_Sugar_BIO	2009	51.1	Existing	3.Biomass_ Ex	8.76	30,000	4.0		40%		1.00
Mayuge_Bio	2015	23.0	Existing	3.Biomass_ Ex	10.48	30,000	4.0		40%		1.00

GM_Sugar_BIO	2021	7.4	Existing	3.Biomass_Ex	12.50	30,000	4.0		40%		1.00
Kamuli_Sugar_BIO	2021	3.0	Existing	3.Biomass_Ex	12.50	30,000	4.0		40%		1.00
Bujagali_HPP	2012	259.0	Existing	7.RoR_Ex		34,313	1.0				1.00
Eskom_HPP	1954	380.0	Existing	7.RoR_Ex		16,906	1.0			32%	1.00
Isimba_HPP	2019	186.0	Existing	7.RoR_Ex		60,813	1.0				1.00
Achwa1_HPP	2021	46.2	Existing	7.RoR_Ex		40,000	1.0				1.00
Achwa2_HPP	2019	41.0	Existing	7.RoR_Ex		106,621	1.0				1.00
Karum_HPP	2021	600.0	Existing	7.RoR_Ex		33,333	1.1			50%	1.00
GOTNUR_BIO	2022	0.4	Existing	3.Biomass_Ex	12.50	30,000	4.0		40%		1.00
Kinyara_Sugar_BIO	2010	14.5	Existing	3.Biomass_Ex	16.88	30,000	4.0		40%		1.00
Hoima_Sugar_BIO	2023	12.0	Existing	3.Biomass_Ex	12.50	30,000	4.0		40%		1.00
Bugoye_HPP	2009	14.0	Existing	7.RoR_Ex		162,776	1.0				1.00
HydromaxHPP	2012	10.0	Existing	7.RoR_Ex		56,865	1.0				1.00
Ishasha_HPP	2011	6.0	Existing	7.RoR_Ex		133,879	1.0				1.00
Kakaka_HPP	2021	4.8	Existing	7.RoR_Ex		136,932	1.0				1.00
Mubukuli_HPP	2008	9.9	Existing	7.RoR_Ex		100,763	1.0				1.00
Kikagati_HPP	2021	14.0	Existing	7.RoR_Ex		551,220	1.0				1.00
Kilembe_Mubukul_HPP	1956	5.0	Existing	7.RoR_Ex		71,405	1.0				1.00
Kyambura_HPP	2019	7.6	Existing	7.RoR_Ex		53,753	1.0				1.00
Lubilia_HPP	2018	5.5	Existing	7.RoR_Ex		104,530	1.0				1.00
Mahoma_HPP	2018	3.6	Existing	7.RoR_Ex		130,321	1.0				1.00
Mpanga_HPP	2011	19.0	Existing	7.RoR_Ex		274,006	1.0				1.00
Muvumbe_HPP	2017	6.6	Existing	7.RoR_Ex		82,756	1.0				1.00
Ndugutu_HPP	2019	6.2	Existing	7.RoR_Ex		86,918	1.0				1.00
Nkusi_HPP	2018	11.6	Existing	7.RoR_Ex		32,656	1.0				1.00
Nyamwamba_HPP	2018	10.0	Existing	7.RoR_Ex		499,211	1.0				1.00
Nyamwambali_HPP	2022	8.1	Existing	7.RoR_Ex		233,816	1.0				1.00
Rwimi_HPP	2017	9.6	Existing	7.RoR_Ex		209,856	1.0				1.00
Sindila_HPP	2019	5.3	Existing	7.RoR_Ex		74,485	1.0				1.00
Timex_Bukinda_HPP	2020	6.7	Existing	7.RoR_Ex		72,066	1.0				1.00
Waki_HPP	2018	5.3	Existing	7.RoR_Ex		65,936	1.0				1.00
Nyamagasani1_HPP	2018	15.0	Existing	7.RoR_Ex		56,022	1.0				1.00
Nyamagasani2_HPP	2021	6.1	Existing	7.RoR_Ex		101,747	1.0				1.00
Bwindi	2014	0.1	Existing	7.RoR_Ex		40,000	1.0				1.00
Kisiizi_Hospital_HPP	2009	0.4	Existing	7.RoR_Ex		40,000	1.0				1.00

Nyagaki_HPP	2012	3.5	Existing	7.RoR_Ex		40,000	1.0				1.00
CENTRAL_ROR_Com	2025	1.0	Committed	8.RoR		40,000	1.0	2.83		40%	1.00
SCOUL_Bagasse_BIO	2025	9.5	Committed	4.Biomass	12.50	75,000	2.5	3.75	40%	5%	1.00
EAST_ROR_Com	2025	27.7	Committed	8.RoR		40,000	1.0	2.83		40%	1.00
Rupa_WPP	2025	20.0	Committed	10.Wind		25,000	1.0	1.30			0.25
WEST_ROR_Com	2025	15.8	Committed	8.RoR		40,000	1.0	2.83		40%	1.00
Kingfisher_Gas_TPP	2025	39.1	Committed	6.OCGT	9.45	20,000	3.5	0.80		10%	1.00
Tilenga_Gas_TPP	2025	200.0	Committed	5.CCGT	6.40	25,000	2.0	0.80	40%	20%	1.00
WestNile_ROR_Com	2025	5.6	Committed	8.RoR		40,000	1.0	2.83		40%	1.00
CENTRAL_Uenergy_BIO	2026	20.0	Candidate	4.Biomass	12.50	75,000	2.5	3.75	40%	5%	1.00
CENTRAL_BIO	2028	200.0	Candidate	4.Biomass	12.50	75,000	2.5	3.75	40%	5%	1.00
CENTRAL_PV	2030	600.0	Candidate	12.PV		10,000	1.0	0.90			0.03
CENTRAL_Diesel	2027	5000.0	Candidate	14.Diesel	12.00	60,000	4.5	0.50		25%	1.00
CENTRAL_Nakasongola_NPP	2036	7200.0	Candidate	1.Nuclear	11.35	55,560	7.0	4.90	80%	5%	1.00
EAST_Jinja_WTE	2026	2.5	Candidate	4.Biomass	12.00	75,000	4.0	3.00	40%	0%	1.00
EAST_ROR_Cand	2025	55.5	Candidate	8.RoR		40,000	1.0	2.83		40%	1.00
EAST_Karamoja_WIND	2026	120.0	Candidate	10.Wind		40,000	1.0	1.30			0.25
EAST_Buranga_GEO	2028	380.0	Candidate	2.Geother- mal		75,000		3.50	90%		1.00
EAST_CCGT	2030	1250.0	Candidate	5.CCGT	6.40	25,000	2.0	0.90	40%	10%	1.00
EAST_GT	2030	1250.0	Candidate	6.OCGT	9.00	20,000	3.5	0.80		20%	1.00
EAST_PV	2030	5800.0	Candidate	12.PV		10,000	1.0	0.90			0.03
EAST_Buyende_NPP	2032	8400.0	Candidate	1.Nuclear	11.35	55,560	7.0	4.90	80%	5%	1.00
Kamuli_Sugar_Bagasse_BIO	2026	3.0	Candidate	4.Biomass	12.50	75,000	2.5	3.75	40%	5%	1.00
Pramukh_Bagasse_BIO	2007	8.0	Candidate	4.Biomass	12.50	75,000	2.5	3.75	40%	5%	1.00
JINJA_Floating_PV	2030	270.0	Candidate	12.PV		10,000	1.0	0.90			0.03
NORTH_Peche_BIO	2026	20.0	Candidate	4.Biomass	12.50	75,000	2.5	3.75	40%	5%	1.00
NORTH_PV	2030	1150.0	Candidate	12.PV		10,000	1.0	0.90			0.03
NORTH_ROR	2026	1412.0	Candidate	8.RoR		40,000	1.0	2.83		40%	1.00
NORTH_Lamor_NPP	2046	4000.0	Candidate	1.Nuclear	11.35	55,560	7.0	4.90	80%	5%	1.00
Kabale_Peat_BIO	2027	30.0	Candidate	4.Biomass	12.50	75,000	2.5	1.70	40%	5%	1.00
Kinyara_Sugar_BIO_Cand	2027	30.0	Candidate	4.Biomass	12.50	75,000	2.5	1.80	40%	0%	1.00
Kyoga_Peat_BIO	2027	50.0	Candidate	4.Biomass	12.50	75,000	2.5	1.70	40%	5%	1.00
WEST_BIOMASS	2028	1120.0	Candidate	4.Biomass	12.50	75,000	2.5	3.75	40%	5%	1.00
WEST_ROR	2026	384.0	Candidate	8.RoR		40,000	1.0	2.83		40%	1.00
WEST_PV	2030	437.0	Candidate	12.PV		10,000	1.0	0.90			0.03
WEST_Katwe_GEO	2026	680.0	Candidate	2.Geother- mal		75,000		3.50	90%		1.00
WEST_CCGT	2030	1500.0	Candidate	5.CCGT	6.40	25,000	2.0	0.90	40%	10%	1.00

WEST_GT	2030	1646.0	Candidate	6.OCGT	9.00	20,000	3.5	0.80		20%	1.00
WEST_Kiruhura_NPP	2041	4800.0	Candidate	1.Nuclear	11.35	55,560	7.0	4.90	80%	5%	1.00
Wind_Generic_West_Nile	2026	80.0	Candidate	10.Wind		40,000	1.0	1.30			0.25
WNILE_Panyimur_GEO	2026	150.0	Candidate	2.Geothe-rmal		75,000		3.50	90%		1.00
WNILE_ROR	2026	800.0	Candidate	8.RoR		40,000	1.0	2.83		40%	1.00
Wnile_PV	2030	1610.0	Candidate	12.PV		10,000	1.0	0.90			0.03
BESS_CENTRAL_2030	2041	750.0	Candidate	15.BESS				0.31		50%	1.00
BESS_CENTRAL_2035	2035	750.0	Candidate	15.BESS				0.31		50%	1.00
BESS_CENTRAL_2040	2041	750.0	Candidate	15.BESS				0.30		50%	1.00
BESS_CENTRAL_2045	2041	750.0	Candidate	15.BESS				0.29		50%	1.00
BESS_CENTRAL_2050	2041	750.0	Candidate	15.BESS				0.28		50%	1.00
BESS_EAST_2030	2041	750.0	Candidate	15.BESS				0.31		50%	1.00
BESS_EAST_2035	2035	750.0	Candidate	15.BESS				0.31		50%	1.00
BESS_EAST_2040	2041	750.0	Candidate	15.BESS				0.30		50%	1.00
BESS_EAST_2045	2041	750.0	Candidate	15.BESS				0.29		50%	1.00
BESS_EAST_2050	2041	750.0	Candidate	15.BESS				0.28		50%	1.00
BESS_JM_2030	2041	750.0	Candidate	15.BESS				0.31		50%	1.00
BESS_JM_2035	2041	750.0	Candidate	15.BESS				0.31		50%	1.00
BESS_JM_2040	2041	750.0	Candidate	15.BESS				0.30		50%	1.00
BESS_JM_2045	2041	750.0	Candidate	15.BESS				0.29		50%	1.00
BESS_JM_2050	2041	750.0	Candidate	15.BESS				0.28		50%	1.00
BESS_NORTH_2030	2041	750.0	Candidate	15.BESS				0.31		50%	1.00
BESS_NORTH_2035	2041	750.0	Candidate	15.BESS				0.31		50%	1.00
BESS_NORTH_2040	2041	750.0	Candidate	15.BESS				0.30		50%	1.00
BESS_NORTH_2045	2041	750.0	Candidate	15.BESS				0.29		50%	1.00
BESS_NORTH_2050	2041	750.0	Candidate	15.BESS				0.28		50%	1.00
BESS_WEST_2030	2041	750.0	Candidate	15.BESS				0.31		50%	1.00
BESS_WEST_2035	2041	750.0	Candidate	15.BESS				0.31		50%	1.00
BESS_WEST_2040	2041	750.0	Candidate	15.BESS				0.30		50%	1.00
BESS_WEST_2045	2041	750.0	Candidate	15.BESS				0.29		50%	1.00
BESS_WEST_2050	2041	750.0	Candidate	15.BESS				0.28		50%	1.00
BESS_WNile_2030	2041	750.0	Candidate	15.BESS				0.31		50%	1.00
BESS_WNile_2035	2041	750.0	Candidate	15.BESS				0.31		50%	1.00
BESS_WNile_2040	2041	750.0	Candidate	15.BESS				0.30		50%	1.00
BESS_WNile_2045	2041	750.0	Candidate	15.BESS				0.29		50%	1.00
BESS_WNile_2050	2041	750.0	Candidate	15.BESS				0.28		50%	1.00

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